

sky130_rhbd_tt_1P8_25C.ccs Library

Cell Groups
AND2X1
AND3X1
AO3X1
AOA4X1
AOAI4X1
AOI3X1
BUFX1
DFFQNX1
DFFQX1
DFFRNQNX1
DFFRNQX1
DFFRNX1
DFFSNQNX1
DFFSNQX1
DFFSNRNQX1
DFFSNRNX1
DFFSNX1
DFFX1
DLATCHN
DLATCH
FA
HA
INVX1

MUX2X1
NAND2X1
NAND3X1
NOR2X1
NOR3X1
OR2X1
OR3X1
TIEHI
TIELO
TMRDFFQNX1
TMRDFFQX1
TMRDFFRNQX1
TMRDFFSNQX1
VOTER3X1
VOTERN3X1
XNOR2X1
XOR2X1

AND2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	x	0
1	0	0
1	1	1

Footprint

Cell Name	Area
AND2X1	50.67280

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
AND2X1	0.01067	0.01080	6.00091

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AND2X1	0.00000	7.37298	13.28970

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND2X1	A->Y (RR)	0.07203	0.64177	7.36437
	B->Y (RR)	0.06540	0.64448	7.38134

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND2X1	A->Y (FF)	0.06135	0.55710	6.28215
	B->Y (FF)	0.05905	0.55738	6.09628

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND2X1	A	0.00000	0.00000	0.00000
	A	8.25196	8.26519	8.56158
	B	0.00000	0.00000	0.00000
	B	8.25215	8.26819	8.57161

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND2X1	A	0.00000	0.00000	0.00000
	A	2.39228	2.41254	2.76296
	B	0.00000	0.00000	0.00000
	B	2.38394	2.39972	2.67165

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.55805	2.55668	2.55761

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	3.15047	3.14985	3.14992

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	2.56109	2.56056	2.56098

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND2X1	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	3.15111	3.15111	3.15158

AND3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	0
1	0	x	0
1	1	0	0
1	1	1	1

Footprint

Cell Name	Area
AND3X1	62.15760

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
AND3X1	0.01068	0.01048	0.01071	6.07005

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AND3X1	0.00000	6.97639	20.76320

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND3X1	A->Y (RR)	0.10049	0.68094	7.62950
	B->Y (RR)	0.09372	0.68879	7.65694
	C->Y (RR)	0.08470	0.69434	7.72042

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AND3X1	A->Y (FF)	0.06852	0.57683	6.35470
	B->Y (FF)	0.06658	0.57111	6.08847
	C->Y (FF)	0.06294	0.57160	6.07302

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND3X1	A	0.00000	0.00000	0.00000
	A	13.03900	13.04930	13.32060
	B	0.00000	0.00000	0.00000
	B	13.04070	13.05010	13.31820
	C	0.00000	0.00000	0.00000
	C	13.04030	13.05200	13.32360

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AND3X1	A	0.00000	0.00000	0.00000
	A	1.66287	1.68257	2.02986
	B	0.00000	0.00000	0.00000
	B	1.65451	1.67016	1.95104
	C	0.00000	0.00000	0.00000
	C	1.64684	1.65946	1.88620

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	2.22209	2.22143	2.22118
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	2.22414	2.22374	2.22358
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.22049	2.22029	2.22025

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	2.78052	2.77971	2.77988
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	2.78182	2.78121	2.78119
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.77891	2.77863	2.77896

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	2.22205	2.22189	2.22180
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	2.22737	2.22681	2.22654
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.22067	2.21975	2.22057

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	2.78014	2.77931	2.77941
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	2.78211	2.78242	2.78225
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.78024	2.78010	2.78056

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.22510	2.22446	2.22459
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	2.22803	2.22635	2.22769

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AND3X1	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.78093	2.78145	2.78109
	(!A * B * !Y)	0.00000	0.00000	0.00000
	(!A * B * !Y)	2.78042	2.78024	2.78035

AO3X1

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Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	1
1	0	x	1
1	1	0	0
1	1	1	1

Footprint

Cell Name	Area
AO3X1	76.51360

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
AO3X1	0.01029	0.01056	0.01056	5.69722

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AO3X1	0.00000	5.28280	19.23130

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AO3X1	A->Y (FR)	0.10450	0.74741	7.65060
	B->Y (FR)	0.10070	0.72235	7.30907
	C->Y (RR)	0.05424	0.61531	6.79931

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AO3X1	A->Y (RF)	0.14547	0.59408	5.04974
	B->Y (RF)	0.13754	0.58431	4.95047
	C->Y (FF)	0.08158	0.60116	6.18054

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AO3X1	A	0.00000	0.00000	0.00000
	A	1.49675	1.51437	1.77617
	B	0.00000	0.00000	0.00000
	B	1.48717	1.50213	1.72737
	C	0.00000	0.00000	0.00000
	C	11.37970	11.39050	11.61540

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AO3X1	A	0.00000	0.00000	0.00000
	A	10.20570	10.21510	10.40070
	B	0.00000	0.00000	0.00000
	B	10.20870	10.21850	10.41830
	C	0.00000	0.00000	0.00000
	C	11.39400	11.40810	11.63140

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	11.81730	11.82600	12.06010
	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.03478	0.03473	0.03471

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.06921	0.08461	0.34459
	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.05831	0.05791	0.05787

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	11.81690	11.82770	12.06570
	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.03739	0.03735	0.03734

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.06040	0.07281	0.29916
	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.05813	0.05834	0.05828

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	(A * !B * Y) + (!A * Y)	0.00000	0.00000	0.00000
	(A * !B * Y) + (!A * Y)	0.02396	0.02396	0.02393

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AO3X1	$(A * !B * Y) + (!A * Y)$	0.00000	0.00000	0.00000
	$(A * !B * Y) + (!A * Y)$	1.47398	1.47384	1.47397

AOA4X1

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1.80, Temp 25.00*

Truth Table

INPUT				OUTPUT
A	B	C	D	Y
0	x	x	x	0
1	0	x	x	0
1	1	x	0	0
1	1	0	1	1
1	1	1	1	0

Footprint

Cell Name	Area
AOA4X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A	B	C	D	Y
AOA4X1	0.01028	0.01052	0.01048	0.01054	5.51490

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOA4X1	0.00000	10.36610	25.24350

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOA4X1	A->Y (RR)	0.20053	0.77182	6.77454
	B->Y (RR)	0.19270	0.77663	6.87993
	C->Y (FR)	0.13556	0.77790	7.69466
	D->Y (RR)	0.06820	0.64887	7.01083

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOA4X1	A->Y (FF)	0.14325	0.62490	5.39139
	B->Y (FF)	0.13966	0.61482	5.27506
	C->Y (RF)	0.09277	0.50269	4.34219
	D->Y (FF)	0.06018	0.55062	5.68179

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOA4X1	A	0.00000	0.00000	0.00000
	A	14.30670	14.31320	14.53890
	B	0.00000	0.00000	0.00000
	B	14.30970	14.31510	14.58980
	C	0.00000	0.00000	0.00000
	C	15.16120	15.17190	15.37410
	D	0.00000	0.00000	0.00000
	D	15.24190	15.25320	15.50100

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOA4X1	A	0.00000	0.00000	0.00000
	A	4.88096	4.89676	5.19597
	B	0.00000	0.00000	0.00000
	B	4.87271	4.88601	5.14851
	C	0.00000	0.00000	0.00000
	C	14.49930	14.50810	14.70950
	D	0.00000	0.00000	0.00000
	D	9.30778	9.32218	9.55464

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	15.38090	15.38900	15.61830
	(B * !C * !D * !Y)	0.00000	0.00000	0.00000
	(B * !C * !D * !Y)	8.11401	8.11804	8.30200
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	2.75283	2.75250	2.75269

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	2.51135	2.52667	2.77991
	(B * !C * !D * !Y)	0.00000	0.00000	0.00000
	(B * !C * !D * !Y)	5.02275	5.03734	5.27783
	(!B * !Y)	0.00000	0.00000	0.00000
	(!B * !Y)	3.06275	3.06246	3.06239

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	(A * C * !Y)	0.00000	0.00000	0.00000
	(A * C * !Y)	15.38060	15.39020	15.62310
	(A * !C * !D * !Y)	0.00000	0.00000	0.00000
	(A * !C * !D * !Y)	8.11986	8.12441	8.31390
	(!A * !Y)	0.00000	0.00000	0.00000
	(!A * !Y)	2.75418	2.75377	2.75437

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(A * C * !Y)$	0.00000	0.00000	0.00000
	$(A * C * !Y)$	2.50235	2.51621	2.74504
	$(A * !C * !D * !Y)$	0.00000	0.00000	0.00000
	$(A * !C * !D * !Y)$	5.01456	5.02710	5.24480
	$(!A * !Y)$	0.00000	0.00000	0.00000
	$(!A * !Y)$	3.06298	3.06301	3.06311

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(A * B * !D * !Y)$	0.00000	0.00000	0.00000
	$(A * B * !D * !Y)$	15.21310	15.22110	15.42700
	$(A * !B * !Y) + (!A * !Y)$	0.00000	0.00000	0.00000
	$(A * !B * !Y) + (!A * !Y)$	1.97982	1.97950	1.97966

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(A * B * !D * !Y)$	0.00000	0.00000	0.00000
	$(A * B * !D * !Y)$	9.30688	9.31705	9.51923
	$(A * !B * !Y) + (!A * !Y)$	0.00000	0.00000	0.00000
	$(A * !B * !Y) + (!A * !Y)$	5.01031	5.00987	5.01020

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)$	0.00000	0.00000	0.00000
	$(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)$	14.46450	14.46370	14.46420

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOA4X1	$(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)$	0.00000	0.00000	0.00000
	$(A * B * C * !Y) + (A * !B * !Y) + (!A * !Y)$	15.10390	15.10400	15.10380

AOAI4X1

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Voltage 1.80, Temp 25.00*

Truth Table

INPUT				OUTPUT
A	B	C	D	YN
0	x	x	x	1
1	0	x	x	1
1	1	x	0	1
1	1	0	1	0
1	1	1	1	1

Footprint

Cell Name	Area
AOAI4X1	85.12720

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	A	B	C	D	YN
AOAI4X1	0.01026	0.01052	0.01053	0.01050	5.27895

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOAI4X1	0.00000	5.05158	26.53990

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOAI4X1	A->YN (FR)	0.11447	0.73432	7.14420
	B->YN (FR)	0.10995	0.70943	6.83771
	C->YN (RR)	0.06315	0.60483	6.32940
	D->YN (FR)	0.03133	0.80024	10.19280

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOAI4X1	A->YN (RF)	0.16359	0.80098	7.49367
	B->YN (RF)	0.15535	0.78270	7.30417
	C->YN (FF)	0.09892	0.80296	8.42497
	D->YN (RF)	0.03319	0.80840	10.27340

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOAI4X1	A	0.00000	0.00000	0.00000
	A	1.33459	1.35033	1.60398
	B	0.00000	0.00000	0.00000
	B	1.32639	1.33993	1.56504
	C	0.00000	0.00000	0.00000
	C	9.89315	9.90275	10.12100
	D	0.00000	0.00000	0.00000
	D	5.16909	5.17144	5.22506

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOAI4X1	A	0.00000	0.00000	0.00000
	A	15.44020	15.44750	15.64040
	B	0.00000	0.00000	0.00000
	B	15.44190	15.44960	15.65430
	C	0.00000	0.00000	0.00000
	C	16.30610	16.31970	16.53030
	D	0.00000	0.00000	0.00000
	D	16.33060	16.33150	16.35100

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * C * YN)	0.00000	0.00000	0.00000
	(B * C * YN)	10.88340	10.89160	11.11590
	(B * !C * !D * YN)	0.00000	0.00000	0.00000
	(B * !C * !D * YN)	4.22290	4.22719	4.40557
	(!B * YN)	0.00000	0.00000	0.00000
	(!B * YN)	0.03323	0.03320	0.03318

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(B * C * YN)	0.00000	0.00000	0.00000
	(B * C * YN)	0.07058	0.08651	0.35181
	(B * !C * !D * YN)	0.00000	0.00000	0.00000
	(B * !C * !D * YN)	1.41044	1.42562	1.66445
	(!B * YN)	0.00000	0.00000	0.00000
	(!B * YN)	0.05453	0.05410	0.05409

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * C * YN)	0.00000	0.00000	0.00000
	(A * C * YN)	10.88320	10.89290	11.12360
	(A * !C * !D * YN)	0.00000	0.00000	0.00000
	(A * !C * !D * YN)	4.22826	4.23294	4.41882
	(!A * YN)	0.00000	0.00000	0.00000
	(!A * YN)	0.03524	0.03520	0.03524

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * C * YN)	0.00000	0.00000	0.00000
	(A * C * YN)	0.06236	0.07530	0.30869
	(A * !C * !D * YN)	0.00000	0.00000	0.00000
	(A * !C * !D * YN)	1.40219	1.41470	1.63084
	(!A * YN)	0.00000	0.00000	0.00000
	(!A * YN)	0.05446	0.05466	0.05460

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * !D * YN)	0.00000	0.00000	0.00000
	(A * B * !D * YN)	10.82340	10.83160	11.03900
	(A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * !B * YN) + (!A * YN)	0.02648	0.02641	0.02644

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * !D * YN)	0.00000	0.00000	0.00000
	(A * B * !D * YN)	5.42379	5.43311	5.64020
	(A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * !B * YN) + (!A * YN)	1.37278	1.37261	1.37285

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	0.00000	0.00000	0.00000
	(A * B * C * YN) + (A * !B * YN) + (!A * YN)	10.13010	10.12860	10.13020

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOAI4X1	$(A * B * C * YN) + (A * !B * YN) + (!A * YN)$	0.00000	0.00000	0.00000
	$(A * B * C * YN) + (A * !B * YN) + (!A * YN)$	10.75620	10.75630	10.75620

AOI3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	YN
0	x	x	0
1	0	x	0
1	1	0	1
1	1	1	0

Footprint

Cell Name	Area
AOI3X1	59.28640

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	YN
AOI3X1	0.01037	0.01062	0.01049	2.87215

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
AOI3X1	0.00000	4.61712	20.74590

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOI3X1	A->YN (RR)	0.10739	0.84204	7.14517
	B->YN (RR)	0.10134	0.85052	7.22976
	C->YN (FR)	0.04590	0.92226	9.98232

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
AOI3X1	A->YN (FF)	0.07139	0.45498	3.79060
	B->YN (FF)	0.06828	0.44417	3.53513
	C->YN (RF)	0.02432	0.51034	5.57924

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOI3X1	A	0.00000	0.00000	0.00000
	A	5.57894	5.58970	5.82295
	B	0.00000	0.00000	0.00000
	B	5.58166	5.59227	5.83221
	C	0.00000	0.00000	0.00000
	C	6.78400	6.78613	6.84462

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
AOI3X1	A	0.00000	0.00000	0.00000
	A	1.63351	1.65196	1.96914
	B	0.00000	0.00000	0.00000
	B	1.62524	1.63937	1.89198
	C	0.00000	0.00000	0.00000
	C	12.73870	12.74060	12.77190

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	13.00800	13.01780	13.27220
	(!B * !YN)	0.00000	0.00000	0.00000
	(!B * !YN)	0.03502	0.03498	0.03496

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	0.06339	0.07754	0.32219
	(!B * !YN)	0.00000	0.00000	0.00000
	(!B * !YN)	0.06079	0.06037	0.06034

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	13.00730	13.01870	13.27270
	(!A * !YN)	0.00000	0.00000	0.00000
	(!A * !YN)	0.03699	0.03692	0.03691

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	0.05508	0.06619	0.27513
	(!A * !YN)	0.00000	0.00000	0.00000
	(!A * !YN)	0.06089	0.06111	0.06105

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * !B * !YN) + (!A * !YN)	0.00000	0.00000	0.00000
	(A * !B * !YN) + (!A * !YN)	0.01823	0.01820	0.01818

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
AOI3X1	(A * !B * !YN) + (!A * !YN)	0.00000	0.00000	0.00000
	(A * !B * !YN) + (!A * !YN)	1.60725	1.60658	1.60685

BUFX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT	OUTPUT
A	Y
0	0
1	1

Footprint

Cell Name	Area
BUFX1	42.05920

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
BUFX1	0.01068	5.86612

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
BUFX1	0.00000	6.07079	6.07079

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
BUFX1	A->Y (RR)	0.05008	0.59196	6.93665

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
BUFX1	A->Y (FF)	0.05439	0.53648	6.11656

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
BUFX1	A	0.00000	0.00000	0.00000
	A	3.52100	3.54041	3.87677

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
BUFX1	A	0.00000	0.00000	0.00000
	A	3.53735	3.55978	3.91281

DFFQNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	CLK	QN
0	R	1
1	R	0
x	x	IQN

Footprint

Cell Name	Area
DFFQNX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	QN
DFFQNX1	0.01055	0.02311	5.42591

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFQNX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQNX1	CLK->QN (RR)	0.15837	0.77417	7.17577

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQNX1	CLK->QN (RF)	0.17648	0.87409	7.85531

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQNX1	hold	CLK (R)	0.04829	0.08078	0.77016
	setup	CLK (R)	0.10103	0.15958	0.75132

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQNX1	hold	CLK (R)	-0.03360	-0.09088	-0.73974
	setup	CLK (R)	0.06400	0.13347	1.56468

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQNX1	min_pulse_width	CLK ()	D	0.10880	0.67383	16.50020
	min_pulse_width	CLK ()	!D	0.14411	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQNX1	min_pulse_width	CLK ()	D	0.18447	0.67383	16.50020
	min_pulse_width	CLK ()	!D	0.09618	0.67383	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	26.25610	26.26540	26.61580

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	21.13080	21.13500	21.34910

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	24.93620	24.93570	24.93790
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	21.09450	21.10080	21.31620
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.29620	11.30260	11.51450

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	25.56960	25.56910	25.56920
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	10.10310	10.11410	10.34910
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.30920	11.31840	11.53790

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	21.09710	21.09810	21.30850
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	26.21420	26.21850	26.41830

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	10.32600	10.33370	10.54020
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	10.11670	10.12290	10.35220
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	9.90653	9.91076	10.11400
	(!D * !QN)	0.00000	0.00000	0.00000
	(!D * !QN)	11.32110	11.32990	11.55020

DFFQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	CLK	Q
0	R	0
1	R	1
x	x	IQ

Footprint

Cell Name	Area
DFFQX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	Q
DFFQX1	0.01057	0.02311	5.42070

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFQX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQX1	CLK->Q (RR)	0.13121	0.77423	7.17777

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFQX1	CLK->Q (RF)	0.20890	0.88167	7.84306

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQX1	hold	CLK (R)	0.04700	0.07678	0.72825
	setup	CLK (R)	0.10466	0.17314	0.78623

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFQX1	hold	CLK (R)	-0.03505	-0.08688	-0.73868
	setup	CLK (R)	0.05834	0.12790	1.53302

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQX1	min_pulse_width	CLK ()	D	0.11636	0.67383	16.50020
	min_pulse_width	CLK ()	!D	0.13654	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFQX1	min_pulse_width	CLK ()	D	0.18951	0.67383	16.50020
	min_pulse_width	CLK ()	!D	0.09366	0.67383	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	21.13070	21.14190	21.49430

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	26.25710	26.26430	26.46880

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	21.09440	21.10070	21.31600
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	24.93620	24.93570	24.93790
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.29260	11.30010	11.50830

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	10.10310	10.11420	10.34910
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	25.56960	25.56910	25.56920
	!CLK	0.00000	0.00000	0.00000
	!CLK	11.30720	11.31640	11.53250

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	21.09640	21.09920	21.30850
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	26.21540	26.21710	26.41790

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	10.11810	10.12520	10.35450
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	10.32590	10.33340	10.54000
	(!D * Q)	0.00000	0.00000	0.00000
	(!D * Q)	11.31930	11.32920	11.54880
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	9.90861	9.91353	10.11610

DFFRNQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
D	RN	CLK	QN
0	1	R	1
1	1	R	0
x	0	x	1
x	1	x	IQN

Footprint

Cell Name	Area
DFFRNQNX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	QN
DFFRNQNX1	0.01013	0.03124	0.02337	3.68547

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNQNX1	0.00000	33.26030	56.43610

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQNX1	CLK->QN (RR)	0.17040	0.68557	5.12498
	RN->QN (FR)	0.04940	0.74192	8.48475

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQNX1	CLK->QN (RF)	0.24732	1.01565	7.91075

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	hold	CLK (R)	0.03920	0.06036	0.83446
	setup	CLK (R)	0.13307	0.17543	0.60193

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	hold	CLK (R)	-0.02117	-0.08078	-0.69449
	setup	CLK (R)	0.06926	0.14502	1.88611

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	hold	CLK (R)	RN	0.03920	0.06036	0.83446
	setup	CLK (R)	RN	0.13307	0.17543	0.60193

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	hold	CLK (R)	RN	-0.02117	-0.08078	-0.69449
	setup	CLK (R)	RN	0.06926	0.14502	1.88611

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQNX1	recovery	CLK (R)	0.12450	0.19759	7.03377
	removal	CLK (R)	0.03487	0.04039	0.01789

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	recovery	CLK (R)	D	0.12450	0.19759	7.03377
	removal	CLK (R)	D	0.03487	0.04039	0.01789

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	RN ()	(CLK * D)	0.12141	0.67383	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.09366	0.67383	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.07096	0.67383	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.07096	0.67383	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	CLK ()	(D * RN)	0.16681	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.15168	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQNX1	min_pulse_width	CLK ()	(D * RN)	0.21726	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.10627	0.67383	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	23.78350	23.79150	24.08490
	RN	24.23260	24.24340	24.51410

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	34.51650	34.51870	34.70790

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * RN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * QN)	22.75580	22.75630	22.75800
	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	34.32610	34.32880	34.52350
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	22.73990	22.74350	22.74400
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	18.84260	18.84630	19.03820
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	9.85933	9.85946	9.85929

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * RN * QN)	0.00000	0.00000	0.00000
	(CLK * RN * QN)	23.36910	23.36980	23.37000
	(CLK * RN * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * !QN)	16.53740	16.54460	16.76100
	(CLK * !RN * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * QN)	23.35990	23.36100	23.36130
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	13.88430	13.89290	14.10360
	(!CLK * !RN * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * QN)	10.43040	10.42960	10.42980

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * D * QN)	22.01790	22.01440	22.01630
	(CLK * !D * QN)	0.00000	0.00000	0.00000
	(CLK * !D * QN)	22.00910	22.00820	22.00920
	(!CLK * D * QN)	0.00000	0.00000	0.00000
	(!CLK * D * QN)	13.82000	13.82390	14.02080
	(!CLK * !D * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * QN)	9.22423	9.22411	9.22435

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(CLK * D * QN)	0.00000	0.00000	0.00000
	(CLK * D * QN)	23.85310	23.85400	23.85450
	(CLK * !D * QN)	0.00000	0.00000	0.00000
	(CLK * !D * QN)	23.85260	23.85490	23.85550
	(!CLK * D * QN)	0.00000	0.00000	0.00000
	(!CLK * D * QN)	10.54300	10.55050	10.74640
	(!CLK * !D * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * QN)	10.91450	10.91340	10.91360

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(D * RN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !QN)	34.32880	34.32740	34.51390
	(D * !RN * QN)	0.00000	0.00000	0.00000
	(D * !RN * QN)	23.89420	23.89550	24.08350
	(!D * RN * QN)	0.00000	0.00000	0.00000
	(!D * RN * QN)	23.90680	23.90830	24.10180
	(!D * !RN * QN)	0.00000	0.00000	0.00000
	(!D * !RN * QN)	23.89190	23.89250	24.08350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQNX1	(D * RN * QN)	0.00000	0.00000	0.00000
	(D * RN * QN)	13.81000	13.81470	13.99940
	(D * RN * !QN)	0.00000	0.00000	0.00000
	(D * RN * !QN)	16.55440	16.55990	16.77150
	(D * !RN * QN)	0.00000	0.00000	0.00000
	(D * !RN * QN)	9.21239	9.21621	9.40996
	(!D * RN * QN)	0.00000	0.00000	0.00000
	(!D * RN * QN)	9.21838	9.22363	9.42134
	(!D * RN * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !QN)	13.90180	13.91020	14.13170
	(!D * !RN * QN)	0.00000	0.00000	0.00000
	(!D * !RN * QN)	9.21524	9.21955	9.41698

DFFRNQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process
, Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
D	RN	CLK	Q
0	1	R	0
1	1	R	1
x	0	x	0
x	1	x	IQ

Footprint

Cell Name	Area
DFFRNQX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	Q
DFFRNQX1	0.01013	0.03181	0.02339	5.48148

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNQX1	0.00000	33.26030	56.43610

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQX1	CLK->Q (RR)	0.18169	0.83054	7.43531

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNQX1	CLK->Q (RF)	0.22207	0.88931	7.78349
	RN->Q (FF)	0.13722	0.93020	8.61900

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	hold	CLK (R)	0.03690	0.05432	0.74541
	setup	CLK (R)	0.14108	0.19502	0.62345

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	hold	CLK (R)	-0.02034	-0.08078	-0.69449
	setup	CLK (R)	0.06350	0.13726	1.83280

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	hold	CLK (R)	RN	0.03690	0.05432	0.74541
	setup	CLK (R)	RN	0.14108	0.19502	0.62345

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	hold	CLK (R)	RN	-0.02034	-0.08078	-0.69449
	setup	CLK (R)	RN	0.06350	0.13726	1.83280

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNQX1	recovery	CLK (R)	0.12959	0.21575	7.16794
	removal	CLK (R)	0.03487	0.04039	0.01789

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	recovery	CLK (R)	D	0.12959	0.21575	7.16794
	removal	CLK (R)	D	0.03487	0.04039	0.01789

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	RN ()	(CLK * D)	0.12141	0.67383	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.10123	0.67383	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.06339	0.67383	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.06339	0.67383	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	CLK ()	(D * RN)	0.17438	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.14411	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNQX1	min_pulse_width	CLK ()	(D * RN)	0.22483	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.10375	0.67383	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	34.51590	34.52340	34.83630

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	23.78430	23.79080	23.99020
	RN	-0.01415	-0.67505	-8.87998
	RN	24.23260	24.25630	24.60620

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	34.32610	34.32880	34.52350
	(CLK * RN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * !Q)	22.75580	22.75630	22.75800
	(CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q)	22.73990	22.74350	22.74400
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	14.42930	14.43310	14.61840
	(!CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q)	9.85931	9.85944	9.85927

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	16.53730	16.54460	16.76090
	(CLK * RN * !Q)	0.00000	0.00000	0.00000
	(CLK * RN * !Q)	23.36910	23.36980	23.37000
	(CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q)	23.35990	23.36090	23.36120
	(!CLK * RN)	0.00000	0.00000	0.00000
	(!CLK * RN)	9.99538	10.00340	10.21160
	(!CLK * !RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q)	10.43040	10.42960	10.42980

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * D * !Q)	0.00000	0.00000	0.00000
	(CLK * D * !Q)	22.01790	22.01440	22.01630
	(CLK * !D * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * !Q)	22.00910	22.00820	22.00910
	(!CLK * D * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * !Q)	13.82000	13.82390	14.02080
	(!CLK * !D * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q)	9.22421	9.22408	9.22433

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(CLK * D * !Q)	0.00000	0.00000	0.00000
	(CLK * D * !Q)	23.85310	23.85400	23.85450
	(CLK * !D * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * !Q)	23.85120	23.85490	23.85550
	(!CLK * D * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * !Q)	10.54300	10.55050	10.74610
	(!CLK * !D * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q)	10.91450	10.91340	10.91360

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(D * RN * Q)	0.00000	0.00000	0.00000
	(D * RN * Q)	34.32870	34.32740	34.51390
	(D * !RN * !Q)	0.00000	0.00000	0.00000
	(D * !RN * !Q)	23.89410	23.89550	24.08350
	(!D * RN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * !Q)	23.90680	23.90830	24.10180
	(!D * !RN * !Q)	0.00000	0.00000	0.00000
	(!D * !RN * !Q)	23.89180	23.89250	24.08350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNQX1	(D * RN * Q)	0.00000	0.00000	0.00000
	(D * RN * Q)	16.55430	16.55990	16.77140
	(D * RN * !Q)	0.00000	0.00000	0.00000
	(D * RN * !Q)	13.81000	13.81470	13.99940
	(D * !RN * !Q)	0.00000	0.00000	0.00000
	(D * !RN * !Q)	9.21237	9.21620	9.40994
	(!D * RN * Q)	0.00000	0.00000	0.00000
	(!D * RN * Q)	13.90180	13.91010	14.13170
	(!D * RN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * !Q)	9.21836	9.22360	9.42132
	(!D * !RN * !Q)	0.00000	0.00000	0.00000
	(!D * !RN * !Q)	9.21522	9.21954	9.41697

DFFRNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT	
D	RN	CLK	Q	QN
0	1	R	0	1
1	1	R	1	0
x	0	x	0	1
x	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFRNX1	208.58881

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	D	RN	CLK	Q	QN
DFFRNX1	0.01013	0.03170	0.02337	5.44593	3.68531

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFRNX1	0.00000	33.26030	56.43610

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->Q (RR)	0.18175	0.82825	7.40006
	QN->Q (FR)	0.05078	0.83598	10.46690

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->Q (RF)	0.23925	1.61817	17.47750
	QN->Q (RF)	0.05555	0.79808	9.94309
	RN->Q (FF)	0.11887	1.70870	21.28830

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->QN (RR)	0.17054	0.68468	5.14395
	Q->QN (FR)	0.04772	0.72816	8.39427
	RN->QN (FR)	0.04975	0.74100	8.49577

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFRNX1	CLK->QN (RF)	0.27044	1.69518	15.50860
	Q->QN (RF)	0.06819	0.90573	10.11430

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	hold	CLK (R)	0.03626	0.05404	0.68801
	setup	CLK (R)	0.14128	0.19484	0.62269

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	hold	CLK (R)	-0.02031	-0.08078	-0.69449
	setup	CLK (R)	0.06106	0.13612	1.81253

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	hold	CLK (R)	RN	0.03626	0.05404	0.68801
	setup	CLK (R)	RN	0.14128	0.19484	0.62269

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	hold	CLK (R)	RN	-0.02031	-0.08078	-0.69449
	setup	CLK (R)	RN	0.06106	0.13612	1.81253

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFRNX1	recovery	CLK (R)	0.13130	0.21586	7.19332
	removal	CLK (R)	0.03487	0.04039	0.01789

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	recovery	CLK (R)	D	0.13130	0.21586	7.19332
	removal	CLK (R)	D	0.03487	0.04039	0.01789

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	RN ()	(CLK * D)	0.12141	0.67383	16.50020
	min_pulse_width	RN ()	(CLK * !D)	0.10123	0.67383	16.50020
	min_pulse_width	RN ()	(!CLK * D)	0.07853	0.67383	16.50020
	min_pulse_width	RN ()	(!CLK * !D)	0.07853	0.67383	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	CLK ()	(D * RN)	0.18195	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.15924	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFRNX1	min_pulse_width	CLK ()	(D * RN)	0.22483	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * RN)	0.10375	0.67383	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.25820	17.25920	17.35230

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	11.89210	11.89520	11.99830
	RN	-0.00707	-0.33621	-4.41119
	RN	12.11690	12.12260	12.21910

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	11.89220	11.89480	11.99670
	RN	-0.00707	-0.26598	-2.98509
	RN	12.11680	12.12280	12.22650

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFRNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.25830	17.25950	17.35220

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * Q * !QN)	34.32600	34.32880	34.52350
	(CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * !Q * QN)	22.75580	22.75630	22.75800
	(CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q * QN)	22.73990	22.74350	22.74400
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	18.84250	18.84630	19.03820
	(!CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q * QN)	9.85931	9.85945	9.85928

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * RN * Q * !QN)	16.53730	16.54460	16.76090
	(CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * RN * !Q * QN)	23.36910	23.36980	23.36990
	(CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !RN * !Q * QN)	23.35990	23.36090	23.36120
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * RN * Q * !QN) + (!CLK * RN * !Q * QN)	13.88420	13.89290	14.10360
	(!CLK * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !RN * !Q * QN)	10.43040	10.42960	10.42980

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * !Q * QN)	22.01780	22.01440	22.01630
	(CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * !Q * QN)	22.00910	22.00820	22.00910
	(!CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * !Q * QN)	13.82010	13.82370	14.02080
	(!CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q * QN)	9.22420	9.22409	9.22433

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * !Q * QN)	23.85300	23.85400	23.85450
	(CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * !Q * QN)	23.85110	23.85490	23.85550
	(!CLK * D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * !Q * QN)	10.54290	10.55050	10.74640
	(!CLK * !D * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !Q * QN)	10.91450	10.91340	10.91360

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * Q * !QN)	34.32870	34.32740	34.51390
	(D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * !RN * !Q * QN)	23.89410	23.89550	24.08350
	(!D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * !Q * QN)	23.90680	23.90830	24.10180
	(!D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !RN * !Q * QN)	23.89180	23.89250	24.08350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFRNX1	(D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * Q * !QN)	16.55460	16.56020	16.77180
	(D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * RN * !Q * QN)	13.81000	13.81470	13.99940
	(D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(D * !RN * !Q * QN)	9.21237	9.21620	9.40994
	(!D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * Q * !QN)	13.90190	13.91010	14.13170
	(!D * RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * !Q * QN)	9.21839	9.22361	9.42133
	(!D * !RN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !RN * !Q * QN)	9.21522	9.21954	9.41697

DFFSNQNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
D	SN	CLK	QN
0	1	R	1
1	1	R	0
x	0	x	0
x	1	x	IQN

Footprint

Cell Name	Area
DFFSNQNX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	QN
DFFSNQNX1	0.01015	0.02231	0.02229	5.23349

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNQNX1	0.00000	30.10480	57.62730

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQNX1	CLK->QN (RR)	0.15629	0.76159	6.98607

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQNX1	CLK->QN (RF)	0.17337	0.84627	7.47287
	SN->QN (FF)	0.26159	0.97744	8.39701

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	hold	CLK (R)	0.01407	0.02420	0.89296
	setup	CLK (R)	0.12264	0.18736	1.79224

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	hold	CLK (R)	-0.06919	-0.15146	-1.14770
	setup	CLK (R)	0.10448	0.19681	1.64800

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	hold	CLK (R)	SN	0.01407	0.02420	0.89296
	setup	CLK (R)	SN	0.12264	0.18736	1.79224

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	hold	CLK (R)	SN	-0.06919	-0.15146	-1.14770
	setup	CLK (R)	SN	0.10448	0.19681	1.64800

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQNX1	recovery	CLK (R)	0.03705	0.03639	0.32051
	removal	CLK (R)	-0.01539	-0.01262	-0.12455

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	recovery	CLK (R)	!D	0.03705	0.03639	0.32051
	removal	CLK (R)	!D	-0.01539	-0.01262	-0.12455

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	SN ()	(CLK * D)	0.07348	0.67383	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.07348	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.07096	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.07096	0.67383	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	CLK ()	(D * SN)	0.10627	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.14663	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQNX1	min_pulse_width	CLK ()	(D * SN)	0.19708	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.11384	0.67383	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	35.36220	35.36980	35.70850

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQNX1	CLK	0.00000	0.00000	0.00000
	CLK	19.64090	19.64790	19.86550
	SN	-0.01415	-0.65655	-8.47822
	SN	18.99930	19.02160	19.37070

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * SN * QN)	0.00000	0.00000	0.00000
	(CLK * SN * QN)	33.71200	33.71230	33.71360
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	19.80280	19.80640	20.00790
	(CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * !QN)	19.79900	19.80320	20.00590
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	10.28760	10.29250	10.49350

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * SN * QN)	0.00000	0.00000	0.00000
	(CLK * SN * QN)	34.35970	34.36020	34.36010
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * SN * !QN) + (!CLK * !SN * !QN)	9.55499	9.56390	9.77692
	(CLK * !SN * !QN)	0.00000	0.00000	0.00000
	(CLK * !SN * !QN)	9.55933	9.56853	9.78785
	(!CLK * SN)	0.00000	0.00000	0.00000
	(!CLK * SN)	14.91830	14.92960	15.14000

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * D * !QN)	0.00000	0.00000	0.00000
	(CLK * D * !QN)	18.32430	18.33140	18.33270
	(CLK * !D * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * !QN)	9.86741	9.87298	9.87298
	(!CLK * D * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * !QN)	9.87598	9.87655	9.87852
	(!CLK * !D * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !QN)	14.41100	14.41480	14.60700

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(CLK * D * !QN)	0.00000	0.00000	0.00000
	(CLK * D * !QN)	19.55180	19.55540	19.55620
	(CLK * !D * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * !QN)	11.02580	11.02730	11.02780
	(!CLK * D * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * !QN)	11.02780	11.02760	11.02850
	(!CLK * !D * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * !QN)	3.67786	3.68777	3.89217

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(D * SN * !QN)	0.00000	0.00000	0.00000
	(D * SN * !QN)	19.79480	19.80160	20.01980
	(D * !SN * !QN)	0.00000	0.00000	0.00000
	(D * !SN * !QN)	19.79230	19.79620	20.01260
	(!D * SN * QN)	0.00000	0.00000	0.00000
	(!D * SN * QN)	35.13980	35.14230	35.34240
	(!D * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * !QN)	10.63550	10.65090	10.91040

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQNX1	(D * SN * QN)	0.00000	0.00000	0.00000
	(D * SN * QN)	13.00340	13.00820	13.24300
	(D * SN * !QN)	0.00000	0.00000	0.00000
	(D * SN * !QN)	9.56595	9.58274	9.85093
	(D * !SN * !QN)	0.00000	0.00000	0.00000
	(D * !SN * !QN)	9.56068	9.57316	9.81653
	(!D * SN * QN)	0.00000	0.00000	0.00000
	(!D * SN * QN)	17.39230	17.39730	17.60760
	(!D * SN * !QN)	0.00000	0.00000	0.00000
	(!D * SN * !QN)	14.92460	14.93450	15.17980
	(!D * !SN * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * !QN)	3.48828	3.49873	3.80065

DFFSNQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
D	SN	CLK	Q
0	1	R	0
1	1	R	1
x	0	x	1
x	1	x	IQ

Footprint

Cell Name	Area
DFFSNQX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	Q
DFFSNQX1	0.01015	0.02175	0.02229	3.66982

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNQX1	0.00000	30.10480	57.62730

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQX1	CLK->Q (RR)	0.12641	0.65459	5.03573
	SN->Q (FR)	0.05072	0.74956	8.56161

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNQX1	CLK->Q (RF)	0.23694	0.98752	7.74574

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	hold	CLK (R)	0.01036	0.02020	0.78547
	setup	CLK (R)	0.13170	0.20114	3.18474

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	hold	CLK (R)	-0.06670	-0.15146	-1.15138
	setup	CLK (R)	0.09968	0.18876	1.60972

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	hold	CLK (R)	SN	0.01036	0.02020	0.78547
	setup	CLK (R)	SN	0.13170	0.20114	3.18474

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	hold	CLK (R)	SN	-0.06670	-0.15146	-1.15138
	setup	CLK (R)	SN	0.09968	0.18876	1.60972

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNQX1	recovery	CLK (R)	0.03356	0.03090	4.60947
	removal	CLK (R)	-0.01539	-0.01262	-0.12507

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	recovery	CLK (R)	!D	0.03356	0.03090	4.60947
	removal	CLK (R)	!D	-0.01539	-0.01262	-0.12507

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	SN ()	(CLK * D)	0.07096	0.67383	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.07096	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.07853	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.07600	0.67383	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.11132	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.13906	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.20213	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10627	0.67383	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	19.64040	19.65150	19.96210
	SN	18.99810	19.01320	19.26980

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	35.36360	35.36980	35.57670

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	$(CLK * SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * Q) + (!CLK * !SN * Q)$	19.80260	19.80630	20.00770
	$(CLK * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q)$	33.71200	33.71230	33.71360
	$(CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !SN * Q)$	19.79900	19.80320	20.00590
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	14.27430	14.27890	14.47990

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	$(CLK * SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * Q) + (!CLK * !SN * Q)$	9.55499	9.56389	9.77692
	$(CLK * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q)$	34.35970	34.36030	34.36010
	$(CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !SN * Q)$	9.55933	9.56852	9.78784
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	19.42270	19.43400	19.64740

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	$(CLK * D * Q)$	0.00000	0.00000	0.00000
	$(CLK * D * Q)$	18.32410	18.33140	18.33270
	$(CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(CLK * !D * Q)$	9.86724	9.87298	9.87298
	$(!CLK * D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * D * Q)$	9.87597	9.87644	9.87852
	$(!CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * Q)$	14.41090	14.41470	14.60700

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(CLK * D * Q)	0.00000	0.00000	0.00000
	(CLK * D * Q)	19.55170	19.55540	19.55620
	(CLK * !D * Q)	0.00000	0.00000	0.00000
	(CLK * !D * Q)	11.02570	11.02730	11.02780
	(!CLK * D * Q)	0.00000	0.00000	0.00000
	(!CLK * D * Q)	11.02780	11.02760	11.02850
	(!CLK * !D * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * Q)	3.67731	3.68717	3.89154

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(D * SN * Q)	0.00000	0.00000	0.00000
	(D * SN * Q)	19.79480	19.80160	20.01980
	(D * !SN * Q)	0.00000	0.00000	0.00000
	(D * !SN * Q)	19.79230	19.79620	20.01260
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	35.13990	35.14240	35.34260
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	10.63550	10.65090	10.91040

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNQX1	(D * SN * Q)	0.00000	0.00000	0.00000
	(D * SN * Q)	9.56595	9.58274	9.85092
	(D * SN * !Q)	0.00000	0.00000	0.00000
	(D * SN * !Q)	13.00340	13.00830	13.24300
	(D * !SN * Q)	0.00000	0.00000	0.00000
	(D * !SN * Q)	9.56055	9.57315	9.81652
	(!D * SN * Q)	0.00000	0.00000	0.00000
	(!D * SN * Q)	14.92460	14.93450	15.17980
	(!D * SN * !Q)	0.00000	0.00000	0.00000
	(!D * SN * !Q)	17.39240	17.39730	17.60760
	(!D * !SN * Q)	0.00000	0.00000	0.00000
	(!D * !SN * Q)	3.48828	3.49872	3.80064

DFFSNRNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT
D	RN	SN	CLK	Q
0	1	1	R	0
1	1	1	R	1
x	x	0	x	1
x	0	1	x	0
x	1	1	x	IQ

Footprint

Cell Name	Area
DFFSNRNQX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)
	D	RN	SN	CLK	Q
DFFSNRNQX1	0.01012	0.03423	0.02266	0.02233	3.55003

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNRNQX1	0.00000	32.92060	54.41090

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQX1	CLK->Q (RR)	0.17501	0.70147	5.08583
	SN->Q (FR)	0.05081	0.73823	8.45663

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNQX1	CLK->Q (RF)	0.24926	0.97749	7.38302
	RN->Q (FF)	0.13331	0.91901	8.08296
	SN->Q (RF)	0.07397	0.89707	9.84027

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	hold	CLK (R)	-0.01381	-0.00818	2.04302
	setup	CLK (R)	0.16803	0.21887	0.69235

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	hold	CLK (R)	-0.06230	-0.14136	-1.08791
	setup	CLK (R)	0.10101	0.19259	2.28526

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.01381	-0.00818	2.04302
	setup	CLK (R)	(RN * SN)	0.16803	0.21887	0.69235

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	hold	CLK (R)	(RN * SN)	-0.06230	-0.14136	-1.08791
	setup	CLK (R)	(RN * SN)	0.10101	0.19259	2.28526

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	recovery	CLK (R)	0.15807	0.24502	7.63114
	removal	CLK (R)	-0.01426	-0.02272	-0.06603
	hold	SN (R)	0.02511	0.06311	0.24929
	setup	SN (R)	0.01384	0.03029	0.96096

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	recovery	CLK (R)	(D * SN)	0.15807	0.24502	7.63114
	removal	CLK (R)	(D * SN)	-0.01426	-0.02272	-0.06603
	hold	SN (R)	(CLK * D)	0.00158	-0.02020	-0.10558
	hold	SN (R)	(CLK * !D)	0.00407	-0.02020	-0.10807
	hold	SN (R)	(!CLK * D)	0.02511	0.06311	0.23003
	hold	SN (R)	(!CLK * !D)	0.02511	0.06311	0.24929
	setup	SN (R)	(CLK * D)	0.00833	0.02777	0.55623
	setup	SN (R)	(CLK * !D)	0.00748	0.03029	0.65697
	setup	SN (R)	(!CLK * D)	0.01253	0.01107	0.96096
	setup	SN (R)	(!CLK * !D)	0.01384	0.01274	0.80352

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	RN ()	(CLK * D * SN)	0.11132	0.67383	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.10880	0.67383	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.07348	0.67383	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.07348	0.67383	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNQX1	recovery	CLK (R)	0.02563	0.01484	4.41094
	removal	CLK (R)	0.00262	0.01010	0.01172
	hold	RN (R)	0.04580	0.10855	0.63725
	setup	RN (R)	0.02466	0.01486	0.56892

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	recovery	CLK (R)	(!D * RN)	0.02563	0.01484	4.41094
	removal	CLK (R)	(!D * RN)	0.00262	0.01010	0.01172
	hold	RN (R)	CLK	0.03876	0.10855	0.63725
	hold	RN (R)	(!CLK * D)	0.04580	0.10097	0.57409
	hold	RN (R)	(!CLK * !D)	0.04580	0.10097	0.59277
	setup	RN (R)	CLK	0.02466	0.01486	0.49020
	setup	RN (R)	(!CLK * D)	-0.00054	0.00505	0.56892
	setup	RN (R)	(!CLK * !D)	-0.00454	0.00505	0.16540

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	SN ()	(CLK * D * RN)	0.08609	0.67383	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.08609	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.08609	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.08609	0.67383	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.16933	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.14411	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNQX1	min_pulse_width	CLK ()	(D * RN * SN)	0.23240	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.11636	0.67383	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	32.67180	32.67850	32.94930
	SN	32.59700	32.60950	32.85660

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNQX1	CLK	0.00000	0.00000	0.00000
	CLK	32.50640	32.51290	32.71460
	RN	-0.01415	-0.52015	-5.75100
	RN	33.06280	33.08440	33.42670
	SN	32.67560	32.67910	32.87860

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	$(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)$	32.63840	32.64110	32.83720
	$(CLK * RN * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q)$	31.10880	31.11070	31.11060
	$(CLK * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * !SN * Q)$	32.63690	32.64010	32.83750
	$(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)$	31.10020	31.10060	31.10100
	$(!CLK * RN * SN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN)$	17.56950	17.57300	17.76560
	$(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	17.19870	17.20060	17.20050

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	$(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * Q) + (!CLK * RN * !SN * Q)$	15.79940	15.80810	16.01610
	$(CLK * RN * SN * !Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q)$	31.73520	31.73480	31.73490
	$(CLK * RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * RN * !SN * Q)$	15.80330	15.80960	16.02450
	$(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * !RN * SN * !Q) + (CLK * !RN * !SN * Q)$	31.72850	31.72830	31.72880
	$(!CLK * RN * SN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN)$	17.58310	17.59330	17.80420
	$(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !RN * SN * !Q) + (!CLK * !RN * !SN * Q)$	17.79820	17.79860	17.79850

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q)	30.36430	30.36320	30.36250
	(CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * D * !SN * Q)	32.88920	32.89090	33.27330
	(CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q)	30.35950	30.35640	30.35860
	(CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * !SN * Q)	17.62490	17.62570	17.90980
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	16.91530	16.92000	17.10820
	(!CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * !SN * Q)	14.12290	14.12750	14.46640
	(!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q)	16.50710	16.50430	16.50620
	(!CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !SN * Q)	3.61076	3.61351	3.78112

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q)	32.23890	32.24300	32.24330
	(CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * D * !SN * Q)	4.84055	4.86785	5.46297
	(CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q)	32.24260	32.24440	32.24480
	(CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * !SN * Q)	6.13285	6.14943	6.54180
	(!CLK * D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q)	18.12460	18.13320	18.32370
	(!CLK * D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * !SN * Q)	0.10889	0.13051	0.55824
	(!CLK * !D * SN * !Q)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q)	18.30290	18.30320	18.30280
	(!CLK * !D * !SN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * !SN * Q)	0.08284	0.09300	0.28323

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q)	30.73390	30.73970	30.74070
	(CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q)	16.86270	16.86600	16.86550
	(!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q)	16.87050	16.86660	16.87170
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	17.58720	17.59160	17.78440

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q)	31.98730	31.99680	31.99740
	(CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q)	18.05860	18.06400	18.06460
	(!CLK * D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q)	18.06310	18.06440	18.06470
	(!CLK * !D * RN * Q)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q)	6.06989	6.07951	6.28627

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(D * RN * SN * Q)	0.00000	0.00000	0.00000
	(D * RN * SN * Q)	32.63360	32.63650	32.82960
	(D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q)	32.63300	32.63530	32.82630
	(!RN * SN * !Q)	0.00000	0.00000	0.00000
	(!RN * SN * !Q)	32.44760	32.45020	32.64070
	(!RN * !SN * Q)	0.00000	0.00000	0.00000
	(!RN * !SN * Q)	3.97122	3.97177	4.11832
	(!D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q)	32.45650	32.45900	32.65370
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	18.13590	18.14660	18.39800

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNQX1	(D * RN * SN * Q)	0.00000	0.00000	0.00000
	(D * RN * SN * Q)	15.81330	15.82600	16.06380
	(D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q)	16.61260	16.61640	16.82590
	(D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q)	15.80780	15.81850	16.03970
	(!RN * SN * !Q)	0.00000	0.00000	0.00000
	(!RN * SN * !Q)	16.18960	16.19300	16.39260
	(!RN * !SN * Q)	0.00000	0.00000	0.00000
	(!RN * !SN * Q)	0.08310	0.09133	0.29352
	(!D * RN * SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q)	17.59970	17.60790	17.85610
	(!D * RN * SN * !Q)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q)	16.18980	16.19680	16.40700
	(!D * RN * !SN * Q)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q)	5.64020	5.64996	5.93704

DFFSNRNX1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT				OUTPUT	
D	RN	SN	CLK	Q	QN
0	1	1	R	0	1
1	1	1	R	1	0
x	0	0	x	1	1
x	0	1	x	0	1
x	1	0	x	1	0
x	1	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFSNRNX1	231.55840

Pin Capacitance Information

Cell Name	Pin Cap(pf)				Max Cap(pf)	
	D	RN	SN	CLK	Q	QN
DFFSNRNX1	0.01012	0.03429	0.02266	0.02233	3.55003	3.57679

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNRNX1	0.00000	32.92060	54.41090

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->Q (RR)	0.17505	0.70112	5.08914
	QN->Q (FR)	0.05904	0.73285	8.26726
	SN->Q (FR)	0.05103	0.74162	8.45708

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->Q (RF)	0.26559	1.49895	13.33210
	QN->Q (RF)	0.08655	0.82741	8.72930
	RN->Q (FF)	0.14723	1.58952	16.98700
	SN->Q (RF)	0.07396	0.89698	9.84025

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->QN (RR)	0.16813	0.67436	5.02463
	Q->QN (FR)	0.04777	0.71818	8.24992
	RN->QN (FR)	0.05053	0.74076	8.47312

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNRNX1	CLK->QN (RF)	0.26466	1.65372	15.00370
	Q->QN (RF)	0.06765	0.88568	9.81575
	RN->QN (RF)	0.26940	1.08836	10.29660
	SN->QN (FF)	0.13650	1.65415	17.89340

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	hold	CLK (R)	-0.01292	-0.01013	2.02931
	setup	CLK (R)	0.16807	0.21878	0.69424

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	hold	CLK (R)	-0.06235	-0.14136	-1.08841
	setup	CLK (R)	0.09954	0.19037	2.19333

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	hold	CLK (R)	(RN * SN)	-0.01292	-0.01013	2.02931
	setup	CLK (R)	(RN * SN)	0.16807	0.21878	0.69424

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	hold	CLK (R)	(RN * SN)	-0.06235	-0.14136	-1.08841
	setup	CLK (R)	(RN * SN)	0.09954	0.19037	2.19333

Constraints(ns) for RN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	recovery	CLK (R)	0.15806	0.24505	7.61768
	removal	CLK (R)	-0.01426	-0.02272	-0.06603
	hold	SN (R)	0.03082	0.07826	0.48774
	setup	SN (R)	0.02883	0.04439	1.33832

Constraints(ns) for RN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	recovery	CLK (R)	(D * SN)	0.15806	0.24505	7.61768
	removal	CLK (R)	(D * SN)	-0.01426	-0.02272	-0.06603
	hold	SN (R)	(CLK * D)	0.00158	-0.02020	-0.10558
	hold	SN (R)	(CLK * !D)	0.00407	-0.02020	-0.10807
	hold	SN (R)	(!CLK * D)	0.03082	0.07826	0.47843
	hold	SN (R)	(!CLK * !D)	0.02682	0.07826	0.48774
	setup	SN (R)	(CLK * D)	0.00925	0.02777	0.50138
	setup	SN (R)	(CLK * !D)	0.00748	0.03029	0.59463
	setup	SN (R)	(!CLK * D)	0.02718	0.04439	1.33832
	setup	SN (R)	(!CLK * !D)	0.02883	0.04039	1.05368

Constraints(ns) for RN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	RN ()	(CLK * D * SN)	0.11384	0.67383	16.50020
	min_pulse_width	RN ()	(CLK * !D * SN)	0.11132	0.67383	16.50020
	min_pulse_width	RN ()	(!CLK * D * SN)	0.08609	0.67383	16.50020
	min_pulse_width	RN ()	(!CLK * !D * SN)	0.08609	0.67383	16.50020

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNRNX1	recovery	CLK (R)	0.02483	0.01044	4.08451
	removal	CLK (R)	0.00262	0.01010	0.01222
	hold	RN (R)	0.05782	0.12622	0.68846
	setup	RN (R)	0.02210	0.01704	0.55802

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	recovery	CLK (R)	(!D * RN)	0.02483	0.01044	4.08451
	removal	CLK (R)	(!D * RN)	0.00262	0.01010	0.01222
	hold	RN (R)	CLK	0.03876	0.10855	0.63725
	hold	RN (R)	(!CLK * D)	0.05534	0.12622	0.66467
	hold	RN (R)	(!CLK * !D)	0.05782	0.12622	0.68846
	setup	RN (R)	CLK	0.02210	0.01704	0.48806
	setup	RN (R)	(!CLK * D)	-0.01394	-0.02272	0.55802
	setup	RN (R)	(!CLK * !D)	-0.01342	-0.02524	0.09653

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	SN ()	(CLK * D * RN)	0.08609	0.67383	16.50020
	min_pulse_width	SN ()	(CLK * !D * RN)	0.08609	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * D * RN)	0.09618	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * !D * RN)	0.09618	0.67383	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.17438	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.15924	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNRNX1	min_pulse_width	CLK ()	(D * RN * SN)	0.23240	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * RN * SN)	0.11636	0.67383	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.33610	16.33770	16.43300
	SN	-0.00707	-0.26008	-2.87550
	SN	16.29920	16.30490	16.40190

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.25320	16.25580	16.35680
	RN	-0.00707	-0.26008	-2.87550
	RN	16.53190	16.53790	16.64060
	SN	32.67590	32.67980	32.87880

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.25310	16.25560	16.35660
	RN	-0.00707	-0.26125	-2.89718
	RN	16.53210	16.53790	16.63980

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNRNX1	CLK	0.00000	0.00000	0.00000
	CLK	16.33610	16.33790	16.43370
	RN	32.89150	32.89560	33.17100
	SN	-0.00707	-0.26125	-2.89718
	SN	16.29910	16.30460	16.40090

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	32.63840	32.64110	32.83720
	$(CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q * QN)$	31.10870	31.11060	31.11060
	$(CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * !SN * Q * !QN)$	32.63720	32.64040	32.83790
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	31.10010	31.10060	31.10090
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	17.57180	17.57660	17.77380
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	17.19870	17.20050	17.20050

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * Q * !QN) + (!CLK * RN * !SN * Q * !QN)$	15.79940	15.80820	16.01610
	$(CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * SN * !Q * QN)$	31.73520	31.73480	31.73490
	$(CLK * RN * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * RN * !SN * Q * !QN)$	15.80340	15.80960	16.02450
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !RN * SN * !Q * QN) + (CLK * !RN * !SN * Q * QN)$	31.72870	31.72850	31.72900
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * RN * SN * Q * !QN) + (!CLK * RN * SN * !Q * QN)$	17.58790	17.59780	17.81410
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !RN * SN * !Q * QN) + (!CLK * !RN * !SN * Q * QN)$	17.79810	17.79860	17.79840

Passive power(pJ) for RN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	$(CLK * D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * D * SN * !Q * QN)$	30.36430	30.36320	30.36250
	$(CLK * !D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * !D * SN * !Q * QN)$	30.35950	30.35640	30.35860
	$(!CLK * D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * D * SN * !Q * QN)$	16.91530	16.91970	17.10820
	$(!CLK * !D * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * !D * SN * !Q * QN)$	16.50710	16.50430	16.50610

Passive power(pJ) for RN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRN1	(CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * D * SN * !Q * QN)	32.23880	32.24290	32.24320
	(CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !D * SN * !Q * QN)	32.24260	32.24430	32.24480
	(!CLK * D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * D * SN * !Q * QN)	18.12450	18.13320	18.32360
	(!CLK * !D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * !D * SN * !Q * QN)	18.30290	18.30320	18.30280

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRN1	(CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q * !QN)	30.73400	30.73970	30.74070
	(CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q * !QN)	16.86270	16.86600	16.86550
	(!CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q * !QN)	16.87050	16.86660	16.87170
	(!CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q * !QN)	17.58720	17.59160	17.78440

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * RN * Q * !QN)	31.98740	31.99680	31.99740
	(CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * RN * Q * !QN)	18.05860	18.06400	18.06460
	(!CLK * D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * RN * Q * !QN)	18.06310	18.06440	18.06470
	(!CLK * !D * RN * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * RN * Q * !QN)	6.06970	6.07931	6.28575

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * Q * !QN)	32.63370	32.63650	32.82970
	(D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q * !QN)	32.63300	32.63540	32.82630
	(!RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!RN * SN * !Q * QN)	32.44750	32.45020	32.64070
	(!RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * Q * QN)	3.97120	3.97175	4.11830
	(!D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q * QN)	32.45640	32.45900	32.65360
	(!D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q * !QN)	18.13590	18.14660	18.39800

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNRNX1	(D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * SN * Q * !QN)	15.81310	15.82610	16.06380
	(D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(D * RN * SN * !Q * QN)	16.61210	16.61610	16.82550
	(D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * RN * !SN * Q * !QN)	15.80770	15.81850	16.03970
	(!RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!RN * SN * !Q * QN)	16.18950	16.19300	16.39260
	(!RN * !SN * Q * QN)	0.00000	0.00000	0.00000
	(!RN * !SN * Q * QN)	0.08308	0.09131	0.29350
	(!D * RN * SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * Q * !QN)	17.59970	17.60790	17.85600
	(!D * RN * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * RN * SN * !Q * QN)	16.18940	16.19640	16.40660
	(!D * RN * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * RN * !SN * Q * !QN)	5.64025	5.64998	5.93703

DFFSNX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT	
D	SN	CLK	Q	QN
0	1	R	0	1
1	1	R	1	0
x	0	x	1	0
x	1	x	IQ	IQN

Footprint

Cell Name	Area
DFFSNX1	197.10400

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	D	SN	CLK	Q	QN
DFFSNX1	0.01015	0.02175	0.02229	3.67547	5.23618

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFSNX1	0.00000	30.10480	57.62730

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->Q (RR)	0.12621	0.65408	5.04865
	QN->Q (FR)	0.05849	0.73998	8.43143
	SN->Q (FR)	0.05048	0.74727	8.60439

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->Q (RF)	0.25325	1.52403	13.94630
	QN->Q (RF)	0.08574	0.84188	9.00544

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->QN (RR)	0.15647	0.76050	7.00273
	Q->QN (FR)	0.04448	0.81357	10.18930

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFSNX1	CLK->QN (RF)	0.19121	1.65732	18.10340
	Q->QN (RF)	0.04611	0.80432	10.03750
	SN->QN (FF)	0.11478	1.72646	21.30640

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	hold	CLK (R)	0.01037	0.02020	0.74856
	setup	CLK (R)	0.12880	0.20459	3.33784

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	hold	CLK (R)	-0.06670	-0.15146	-1.15135
	setup	CLK (R)	0.09957	0.18895	1.59473

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	hold	CLK (R)	SN	0.01037	0.02020	0.74856
	setup	CLK (R)	SN	0.12880	0.20459	3.33784

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	hold	CLK (R)	SN	-0.06670	-0.15146	-1.15135
	setup	CLK (R)	SN	0.09957	0.18895	1.59473

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFSNX1	recovery	CLK (R)	0.03331	0.02988	4.32603
	removal	CLK (R)	-0.01690	-0.01262	-0.12507

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	recovery	CLK (R)	!D	0.03331	0.02988	4.32603
	removal	CLK (R)	!D	-0.01690	-0.01262	-0.12507

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	SN ()	(CLK * D)	0.07096	0.67383	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.07096	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.08357	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.08357	0.67383	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	CLK ()	(D * SN)	0.11636	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.15168	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFSNX1	min_pulse_width	CLK ()	(D * SN)	0.20213	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10627	0.67383	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	9.81939	9.82328	9.93528
	SN	-0.00707	-0.26555	-2.97713
	SN	9.50014	9.50567	9.60658

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.68170	17.68410	17.78800

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	17.68140	17.68390	17.78760

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFSNX1	CLK	0.00000	0.00000	0.00000
	CLK	9.82036	9.82379	9.93667
	SN	-0.00707	-0.32838	-4.24129
	SN	9.49981	9.50476	9.59971

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	$(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)$	19.80260	19.80620	20.00770
	$(CLK * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q * QN)$	33.71200	33.71230	33.71360
	$(CLK * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * !SN * Q * !QN)$	19.79890	19.80310	20.00590
	$(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)$	10.28720	10.29200	10.49300

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	$(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * SN * Q * !QN) + (!CLK * !SN * Q * !QN)$	9.55497	9.56387	9.77689
	$(CLK * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(CLK * SN * !Q * QN)$	34.35970	34.36020	34.36010
	$(CLK * !SN * Q * !QN)$	0.00000	0.00000	0.00000
	$(CLK * !SN * Q * !QN)$	9.55931	9.56849	9.78781
	$(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)$	0.00000	0.00000	0.00000
	$(!CLK * SN * Q * !QN) + (!CLK * SN * !Q * QN)$	14.91750	14.92960	15.14000

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * Q * !QN)	18.32370	18.33130	18.33250
	(CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * Q * !QN)	9.86694	9.87330	9.87330
	(!CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * Q * !QN)	9.87569	9.87532	9.87827
	(!CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * Q * !QN)	14.41030	14.41430	14.60650

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * D * Q * !QN)	19.55160	19.55540	19.55620
	(CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * !D * Q * !QN)	11.02530	11.02720	11.02770
	(!CLK * D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * D * Q * !QN)	11.02760	11.02750	11.02780
	(!CLK * !D * Q * !QN)	0.00000	0.00000	0.00000
	(!CLK * !D * Q * !QN)	3.67752	3.68742	3.89183

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * SN * Q * !QN)	19.79470	19.80140	20.01960
	(D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * !SN * Q * !QN)	19.79240	19.79620	20.01260
	(!D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * SN * !Q * QN)	35.14010	35.14250	35.34260
	(!D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * Q * !QN)	10.63560	10.65100	10.91050

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFSNX1	(D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * SN * Q * !QN)	9.56600	9.58281	9.85099
	(D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(D * SN * !Q * QN)	13.00490	13.00970	13.24420
	(D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(D * !SN * Q * !QN)	9.56071	9.57337	9.81662
	(!D * SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * SN * Q * !QN)	14.92470	14.93460	15.17990
	(!D * SN * !Q * QN)	0.00000	0.00000	0.00000
	(!D * SN * !Q * QN)	17.39240	17.39740	17.60770
	(!D * !SN * Q * !QN)	0.00000	0.00000	0.00000
	(!D * !SN * Q * !QN)	3.48846	3.49881	3.80088

DFFX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT		OUTPUT	
D	CLK	Q	QN
0	R	0	1
1	R	1	0
x	x	IQ	IQN

Footprint

Cell Name	Area
DFFX1	174.13440

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	D	CLK	Q	QN
DFFX1	0.01055	0.02310	5.42033	5.42584

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DFFX1	0.00000	28.46920	43.32390

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->Q (RR)	0.13108	0.77300	7.19024
	QN->Q (FR)	0.05077	0.83600	10.45400

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->Q (RF)	0.22699	1.61520	17.57770
	QN->Q (RF)	0.05584	0.79810	9.93421

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->QN (RR)	0.15857	0.77315	7.20227
	Q->QN (FR)	0.04486	0.82601	10.43860

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DFFX1	CLK->QN (RF)	0.19642	1.70944	18.81930
	Q->QN (RF)	0.04721	0.82617	10.40220

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFX1	hold	CLK (R)	0.04570	0.07678	0.67658
	setup	CLK (R)	0.10793	0.17414	0.78554

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
DFFX1	hold	CLK (R)	-0.03442	-0.08688	-0.73868
	setup	CLK (R)	0.05900	0.12317	1.49751

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFX1	min_pulse_width	CLK ()	D	0.12141	0.67383	16.50020
	min_pulse_width	CLK ()	!D	0.14916	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
DFFX1	min_pulse_width	CLK ()	D	0.18951	0.67383	16.50020
	min_pulse_width	CLK ()	!D	0.09366	0.67383	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	10.56510	10.56820	10.67500

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	13.12850	13.13150	13.23750

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	13.12790	13.13130	13.23580

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DFFX1	CLK	0.00000	0.00000	0.00000
	CLK	10.56550	10.56810	10.67660

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(CLK * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * Q * !QN)	21.09460	21.10080	21.31620
	(CLK * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !Q * QN)	24.93620	24.93570	24.93790
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	11.29620	11.30260	11.51450

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(CLK * Q * !QN)	0.00000	0.00000	0.00000
	(CLK * Q * !QN)	10.10310	10.11420	10.34910
	(CLK * !Q * QN)	0.00000	0.00000	0.00000
	(CLK * !Q * QN)	25.56960	25.56900	25.56920
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	0.00000	0.00000	0.00000
	(!CLK * Q * !QN) + (!CLK * !Q * QN)	11.30920	11.31840	11.53790

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(D * Q * !QN)	0.00000	0.00000	0.00000
	(D * Q * !QN)	21.09710	21.09810	21.30850
	(!D * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !Q * QN)	26.21420	26.21850	26.41820

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DFFX1	(D * Q * !QN)	0.00000	0.00000	0.00000
	(D * Q * !QN)	10.11670	10.12290	10.35220
	(D * !Q * QN)	0.00000	0.00000	0.00000
	(D * !Q * QN)	10.32600	10.33360	10.54020
	(!D * Q * !QN)	0.00000	0.00000	0.00000
	(!D * Q * !QN)	11.32110	11.32990	11.55020
	(!D * !Q * QN)	0.00000	0.00000	0.00000
	(!D * !Q * QN)	9.90650	9.91074	10.11400

DLATCHN

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	GATE_N	Q
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
DLATCHN	179.87680

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	GATE_N	Q
DLATCHN	0.02276	0.01036	2.88873

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DLATCHN	0.00000	22.21240	28.71060

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCHN	D->Q (RR)	0.18743	0.93064	6.98398
	GATE_N->Q (-R)	0.23709	1.05991	8.01314

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCHN	D->Q (FF)	0.15991	0.53402	3.42270
	GATE_N->Q (-F)	0.17508	0.57777	3.50583

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCHN	D	0.00000	0.00000	0.00000
	D	13.54490	13.56990	14.09900
	GATE_N	0.00000	0.00000	0.00000
	GATE_N	13.29340	13.30570	13.53950

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCHN	D	0.00000	0.00000	0.00000
	D	9.19449	9.22171	9.74499
	GATE_N	0.00000	0.00000	0.00000
	GATE_N	7.78991	7.79975	8.08816

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	GATE_N	0.00000	0.00000	0.00000
	GATE_N	16.22130	16.23270	16.47220

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	GATE_N	0.00000	0.00000	0.00000
	GATE_N	11.66400	11.67960	11.91190

Passive power(pJ) for GATE_N rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	16.31890	16.32940	16.59530
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	11.95600	11.96650	12.20380

Passive power(pJ) for GATE_N falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCHN	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	11.77310	11.78740	12.03900
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	8.98987	9.00626	9.25036

DLATCH

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	GATE	Q
x	0	1
0	1	0
1	1	1

Footprint

Cell Name	Area
DLATCH	162.64960

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	GATE	Q
DLATCH	0.02279	0.02081	2.90667

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
DLATCH	0.00000	18.14490	24.27320

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCH	D->Q (RR)	0.18684	0.92931	6.99745
	GATE->Q (-R)	0.18615	0.91529	6.90747

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
DLATCH	D->Q (FF)	0.15846	0.53319	3.43223
	GATE->Q (-F)	0.11887	0.44728	2.63781

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCH	D	0.00000	0.00000	0.00000
	D	14.12820	14.15470	14.69420
	GATE	0.00000	0.00000	0.00000
	GATE	13.88830	13.89630	14.14810

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
DLATCH	D	0.00000	0.00000	0.00000
	D	9.55203	9.58022	10.10720
	GATE	0.00000	0.00000	0.00000
	GATE	8.12024	8.12853	8.41170

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	!GATE	0.00000	0.00000	0.00000
	!GATE	12.15620	12.16770	12.40740

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	!GATE	0.00000	0.00000	0.00000
	!GATE	8.01428	8.02772	8.26039

Passive power(pJ) for GATE rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	12.37280	12.38330	12.61050
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	9.32277	9.33342	9.55307

Passive power(pJ) for GATE falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
DLATCH	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	12.18690	12.20270	12.45500
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	8.23740	8.24791	8.47690

FA

sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT	
A	B	CIN	COUT	SUM
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

Footprint

Cell Name	Area
FA	309.08081

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	A	B	CIN	COUT	SUM
FA	0.03793	0.03735	0.03457	5.74192	2.61746

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
FA	0.00000	36.98230	51.95610

Delay Information

Delay(ns) to COUT rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
FA	A->COUT (RR)	0.32490	0.88993	7.02011
	B->COUT (RR)	0.34881	0.91702	7.07676
	CIN->COUT (RR)	0.14786	0.72385	6.91082

Delay(ns) to COUT falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
FA	A->COUT (FF)	0.29894	0.81287	6.00401
	B->COUT (FF)	0.30862	0.82572	6.09275
	CIN->COUT (FF)	0.17142	0.69046	5.84574

Delay(ns) to SUM rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
FA	A->SUM (-R)	-	0.25871	1.11177	8.34626
	B->SUM (-R)	-	0.24133	1.08717	8.11268
	CIN->SUM (RR)	$(A * B) + (!A * !B)$	0.10834	0.81170	6.60626
	CIN->SUM (FR)	$(A * !B) + (!A * B)$	0.07148	0.97295	9.99325

Delay(ns) to SUM falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
FA	A->SUM (-F)	-	0.23344	0.80555	5.31322
	B->SUM (-F)	-	0.21548	0.78279	5.14248
	CIN->SUM (FF)	$(A * B) + (!A * !B)$	0.08517	0.58002	4.52819
	CIN->SUM (RF)	$(A * !B) + (!A * B)$	0.04545	0.66514	7.00185

Power Information

Internal switching power(pJ) to COUT rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
FA	A	17.40010	17.41600	17.77020
	B	17.39970	17.41510	17.77210
	CIN	0.00000	0.00000	0.00000
	CIN	14.35090	14.36150	14.61390

Internal switching power(pJ) to COUT falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
FA	A	5.75171	5.75981	5.87940
	B	5.75504	5.76109	5.87764
	CIN	0.00000	0.00000	0.00000
	CIN	3.75756	3.77095	4.02641

Internal switching power(pJ) to SUM rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
FA	A	-	0.00000	0.00000	0.00000
	A	-	26.85280	26.88890	27.56340
	B	-	0.00000	0.00000	0.00000
	B	-	26.85400	26.88830	27.55930
	CIN	(A * B) + (!A * !B)	0.00000	0.00000	0.00000
	CIN	(A * B) + (!A * !B)	27.04000	27.04960	27.28340
	CIN	(A * !B) + (!A * B)	0.00000	0.00000	0.00000
	CIN	(A * !B) + (!A * B)	3.75649	3.77145	4.03522

Internal switching power(pJ) to SUM falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
FA	A	-	0.00000	0.00000	0.00000
	A	-	28.07870	28.28670	29.73710
	B	-	0.00000	0.00000	0.00000
	B	-	28.07480	28.23260	29.23270
	CIN	$(A * B) + (!A * !B)$	0.00000	0.00000	0.00000
	CIN	$(A * B) + (!A * !B)$	33.67490	33.68800	33.91570
	CIN	$(A * !B) + (!A * B)$	0.00000	0.00000	0.00000
	CIN	$(A * !B) + (!A * B)$	14.35080	14.36190	14.62400

HA

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80, Temp
25.00*

Truth Table

INPUT		OUTPUT	
A	B	COUT	SUM
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

Footprint

Cell Name	Area
HA	136.80881

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	A	B	COUT	SUM
HA	0.03356	0.03772	6.17943	2.60808

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
HA	0.00000	17.03700	32.46250

Delay Information

Delay(ns) to COUT rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
HA	A->COUT (RR)	0.07180	0.66828	7.65971
	B->COUT (RR)	0.06578	0.67275	7.70589

Delay(ns) to COUT falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
HA	A->COUT (FF)	0.06435	0.59109	6.29787
	B->COUT (FF)	0.06079	0.57961	6.25763

Delay(ns) to SUM rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
HA	A->SUM (RR)	!B	0.08593	0.78306	6.48830
	A->SUM (FR)	B	0.07772	0.99125	10.08150
	B->SUM (RR)	!A	0.10817	0.79378	6.42662
	B->SUM (FR)	A	0.06972	0.98065	10.05620

Delay(ns) to SUM falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
HA	A->SUM (FF)	!B	0.07766	0.56648	4.61616
	A->SUM (RF)	B	0.04697	0.63457	6.57545
	B->SUM (FF)	!A	0.08605	0.58854	4.58569
	B->SUM (RF)	A	0.04148	0.65619	6.90808

Power Information

Internal switching power(pJ) to COUT rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
HA	A	0.00000	0.00000	0.00000
	A	11.85110	11.86500	12.13420
	B	0.00000	0.00000	0.00000
	B	11.85090	11.86540	12.14100

Internal switching power(pJ) to COUT falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
HA	A	0.00000	0.00000	0.00000
	A	0.86447	0.88122	1.14894
	B	0.00000	0.00000	0.00000
	B	0.85663	0.87139	1.11641

Internal switching power(pJ) to SUM rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
HA	A	B	0.00000	0.00000	0.00000
	A	B	0.86472	0.88146	1.16340
	A	!B	0.00000	0.00000	0.00000
	A	!B	3.99911	4.01138	4.26147
	B	A	0.00000	0.00000	0.00000
	B	A	0.85673	0.87158	1.12578
	B	!A	0.00000	0.00000	0.00000
	B	!A	4.01497	4.02336	4.28243

Internal switching power(pJ) to SUM falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
HA	A	B	0.00000	0.00000	0.00000
	A	B	11.85070	11.86560	12.15270
	A	!B	0.00000	0.00000	0.00000
	A	!B	9.43950	9.45441	9.69810
	B	A	0.00000	0.00000	0.00000
	B	A	11.85120	11.86580	12.16250
	B	!A	0.00000	0.00000	0.00000
	B	!A	9.44620	9.45930	9.67637

INVX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT	OUTPUT
A	Y
0	1
1	0

Footprint

Cell Name	Area
INVX1	24.83200

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	A	Y
INVX1	0.01040	5.57495

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
INVX1	0.00000	3.90439	7.80687

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
INVX1	A->Y (FR)	0.02681	0.76525	9.91760

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
INVX1	A->Y (RF)	0.02028	0.57312	7.36203

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
INVX1	A	0.00000	0.00000	0.00000
	A	0.01846	0.02358	0.08289

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
INVX1	A	0.00000	0.00000	0.00000
	A	4.60683	4.60859	4.63402

MUX2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A0	A1	S	Y
0	0	x	0
x	1	0	1
0	1	1	0
1	0	0	0
1	x	1	1

Footprint

Cell Name	Area
MUX2X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A0	A1	S	Y
MUX2X1	0.01056	0.01054	0.02177	5.42552

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
MUX2X1	0.00000	13.86620	17.36010

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
MUX2X1	A0->Y (RR)	-	0.07933	0.65367	6.88313
	A1->Y (RR)	-	0.07242	0.64433	6.84105
	S->Y (RR)	(A0 * !A1)	0.08322	0.63911	6.76020
	S->Y (FR)	(!A0 * A1)	0.11541	0.72384	7.19830

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
MUX2X1	A0->Y (FF)	-	0.07879	0.77429	8.29389
	A1->Y (FF)	-	0.07260	0.74818	8.18526
	S->Y (FF)	(A0 * !A1)	0.08134	0.77319	8.25537
	S->Y (RF)	(!A0 * A1)	0.10395	0.68610	6.75608

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
MUX2X1	A0	-	0.00000	0.00000	0.00000
	A0	-	9.83630	9.84639	10.09650
	A1	-	0.00000	0.00000	0.00000
	A1	-	5.46006	5.47116	5.71171
	S	(A0 * !A1)	0.00000	0.00000	0.00000
	S	(A0 * !A1)	10.13190	10.15690	10.69710
	S	(!A0 * A1)	0.00000	0.00000	0.00000
	S	(!A0 * A1)	5.71775	5.73316	5.96213

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
MUX2X1	A0	-	0.00000	0.00000	0.00000
	A0	-	9.85199	9.86528	10.08950
	A1	-	0.00000	0.00000	0.00000
	A1	-	5.47656	5.49103	5.72998
	S	(A0 * !A1)	0.00000	0.00000	0.00000
	S	(A0 * !A1)	5.14876	5.18623	5.74431
	S	(!A0 * A1)	0.00000	0.00000	0.00000
	S	(!A0 * A1)	9.55817	9.56874	9.80113

Passive power(pJ) for A0 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	(A1 * !S * Y) + (!A1 * !S * !Y)	0.00000	0.00000	0.00000
	(A1 * !S * Y) + (!A1 * !S * !Y)	5.12196	5.12214	5.12366

Passive power(pJ) for A0 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	$(A1 * !S * Y) + (!A1 * !S * !Y)$	0.00000	0.00000	0.00000
	$(A1 * !S * Y) + (!A1 * !S * !Y)$	5.70511	5.70544	5.70573

Passive power(pJ) for A1 rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	$(A0 * S * Y) + (!A0 * S * !Y)$	0.00000	0.00000	0.00000
	$(A0 * S * Y) + (!A0 * S * !Y)$	9.47483	9.47266	9.47443

Passive power(pJ) for A1 falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	$(A0 * S * Y) + (!A0 * S * !Y)$	0.00000	0.00000	0.00000
	$(A0 * S * Y) + (!A0 * S * !Y)$	10.08610	10.08610	10.08610

Passive power(pJ) for S rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	$(A0 * A1 * Y)$	0.00000	0.00000	0.00000
	$(A0 * A1 * Y)$	9.89140	9.91419	10.41400
	$(!A0 * !A1 * !Y)$	0.00000	0.00000	0.00000
	$(!A0 * !A1 * !Y)$	9.84615	9.85705	10.09490

Passive power(pJ) for S falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
MUX2X1	(A0 * A1 * Y)	0.00000	0.00000	0.00000
	(A0 * A1 * Y)	5.47602	5.50548	5.99341
	(!A0 * !A1 * !Y)	0.00000	0.00000	0.00000
	(!A0 * !A1 * !Y)	5.42800	5.44295	5.68488

NAND2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	x	1
1	0	1
1	1	0

Footprint

Cell Name	Area
NAND2X1	33.44560

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
NAND2X1	0.01054	0.01061	5.10239

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NAND2X1	0.00000	3.89237	15.56580

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND2X1	A->Y (FR)	0.03446	0.77919	9.79964
	B->Y (FR)	0.03032	0.77148	9.73368

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND2X1	A->Y (RF)	0.03781	0.77327	9.66751
	B->Y (RF)	0.03045	0.74872	9.37707

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND2X1	A	0.00000	0.00000	0.00000
	A	0.02834	0.03205	0.08023
	B	0.00000	0.00000	0.00000
	B	0.01894	0.02266	0.05611

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND2X1	A	0.00000	0.00000	0.00000
	A	9.73366	9.73407	9.74473
	B	0.00000	0.00000	0.00000
	B	9.73371	9.73464	9.74941

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	-0.00180	-0.00186	-0.00180

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.01092	0.01055	0.01036

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.00105	0.00100	0.00100

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND2X1	(!A * Y)	0.00000	0.00000	0.00000
	(!A * Y)	0.01104	0.01118	0.01127

NAND3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	x	x	1
1	0	x	1
1	1	0	1
1	1	1	0

Footprint

Cell Name	Area
NAND3X1	44.93040

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
NAND3X1	0.01059	0.01035	0.01056	3.37535

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NAND3X1	0.00000	2.91632	23.32440

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND3X1	A->Y (FR)	0.04078	0.69521	7.93026
	B->Y (FR)	0.03771	0.68762	7.88655
	C->Y (FR)	0.03292	0.68307	7.88261

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NAND3X1	A->Y (RF)	0.06032	0.84244	9.05898
	B->Y (RF)	0.05293	0.82720	9.05906
	C->Y (RF)	0.04341	0.81670	9.14826

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND3X1	A	0.00000	0.00000	0.00000
	A	0.03790	0.04139	0.09703
	B	0.00000	0.00000	0.00000
	B	0.02907	0.03242	0.08476
	C	0.00000	0.00000	0.00000
	C	0.02034	0.02377	0.07328

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NAND3X1	A	0.00000	0.00000	0.00000
	A	14.71990	14.71930	14.73040
	B	0.00000	0.00000	0.00000
	B	14.72010	14.72050	14.73060
	C	0.00000	0.00000	0.00000
	C	14.72060	14.72050	14.73550

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	-0.00242	-0.00246	-0.00243
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.00067	0.00061	0.00062
	(!B * !C * Y)	0.00000	0.00000	0.00000
	(!B * !C * Y)	-0.00451	-0.00453	-0.00456

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.01127	0.01065	0.01061
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.01272	0.01231	0.01225
	(!B * !C * Y)	0.00000	0.00000	0.00000
	(!B * !C * Y)	0.00996	0.00995	0.00995

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	-0.00212	-0.00217	-0.00219
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.00425	0.00423	0.00423
	(!A * !C * Y)	0.00000	0.00000	0.00000
	(!A * !C * Y)	-0.00467	-0.00470	-0.00473

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	0.01081	0.01030	0.01029
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.01201	0.01215	0.01216
	(!A * !C * Y)	0.00000	0.00000	0.00000
	(!A * !C * Y)	0.01103	0.01116	0.01116

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.00019	0.00012	0.00010
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	0.00448	0.00442	0.00444

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NAND3X1	(!B * Y)	0.00000	0.00000	0.00000
	(!B * Y)	0.01100	0.01120	0.01122
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	0.01079	0.01095	0.01089

NOR2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	1
x	1	0
1	x	0

Footprint

Cell Name	Area
NOR2X1	33.44560

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
NOR2X1	0.01081	0.01051	2.54971

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NOR2X1	0.00000	2.00110	6.04921

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR2X1	A->Y (FR)	0.06136	0.94705	9.66809
	B->Y (FR)	0.04503	0.85538	9.03151

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR2X1	A->Y (RF)	0.02615	0.47359	4.90216
	B->Y (RF)	0.02317	0.46680	4.87768

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR2X1	A	0.00000	0.00000	0.00000
	A	0.03271	0.03368	0.06680
	B	0.00000	0.00000	0.00000
	B	0.02727	0.03133	0.09585

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR2X1	A	0.00000	0.00000	0.00000
	A	2.45444	2.45600	2.48863
	B	0.00000	0.00000	0.00000
	B	2.72039	2.71927	2.74573

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	0.06816	0.06684	0.06652

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	3.67749	3.67705	3.67695

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(A * !Y)	0.00000	0.00000	0.00000
	(A * !Y)	0.01664	0.01663	0.01663

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR2X1	(A * !Y)	0.00000	0.00000	0.00000
	(A * !Y)	2.45798	2.45770	2.45805

NOR3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	0	0	1
0	x	1	0
x	1	x	0
1	x	x	0

Footprint

Cell Name	Area
NOR3X1	44.93040

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
NOR3X1	0.01088	0.01045	0.01046	1.64135

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
NOR3X1	0.00000	1.10717	5.39923

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR3X1	A->Y (FR)	0.10649	1.09387	9.36207
	B->Y (FR)	0.09207	1.04313	9.23871
	C->Y (FR)	0.06731	0.95784	8.83971

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
NOR3X1	A->Y (RF)	0.02829	0.42360	3.82958
	B->Y (RF)	0.02840	0.41854	3.78834
	C->Y (RF)	0.02470	0.41098	3.76821

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR3X1	A	0.00000	0.00000	0.00000
	A	0.04650	0.04673	0.07755
	B	0.00000	0.00000	0.00000
	B	0.04113	0.04173	0.07645
	C	0.00000	0.00000	0.00000
	C	0.03609	0.03956	0.10304

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
NOR3X1	A	0.00000	0.00000	0.00000
	A	2.50884	2.51008	2.56158
	B	0.00000	0.00000	0.00000
	B	1.81905	1.82041	1.85455
	C	0.00000	0.00000	0.00000
	C	1.87878	1.87942	1.90831

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	0.01877	0.01809	0.01794
	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	0.09025	0.08895	0.08870
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	0.06609	0.06442	0.06414

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(B * C * !Y)	0.00000	0.00000	0.00000
	(B * C * !Y)	0.06879	0.06877	0.06888
	(B * !C * !Y)	0.00000	0.00000	0.00000
	(B * !C * !Y)	1.79562	1.79569	1.79555
	(!B * C * !Y)	0.00000	0.00000	0.00000
	(!B * C * !Y)	3.07999	3.07953	3.07965

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(A * C * !Y)	0.00000	0.00000	0.00000
	(A * C * !Y)	0.02223	0.02222	0.02226
	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	0.04643	0.04641	0.04642
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	0.04421	0.04258	0.04224

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(A * C * !Y)	0.00000	0.00000	0.00000
	(A * C * !Y)	0.07953	0.07955	0.07959
	(A * !C * !Y)	0.00000	0.00000	0.00000
	(A * !C * !Y)	2.50836	2.50813	2.50866
	(!A * C * !Y)	0.00000	0.00000	0.00000
	(!A * C * !Y)	3.11777	3.11703	3.11718

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	0.00825	0.00824	0.00824
	(A * !B * !Y)	0.00000	0.00000	0.00000
	(A * !B * !Y)	0.01213	0.01213	0.01213

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
NOR3X1	(B * !Y)	0.00000	0.00000	0.00000
	(B * !Y)	0.11687	0.11686	0.11689
	(A * !B * !Y)	0.00000	0.00000	0.00000
	(A * !B * !Y)	2.50985	2.50957	2.50994

OR2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	0
x	1	1
1	x	1

Footprint

Cell Name	Area
OR2X1	50.67280

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
OR2X1	0.01076	0.01066	5.70062

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
OR2X1	0.00000	3.08382	5.40294

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR2X1	A->Y (RR)	0.05627	0.57672	6.45627
	B->Y (RR)	0.05339	0.60440	6.74836

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR2X1	A->Y (FF)	0.09491	0.59873	6.23996
	B->Y (FF)	0.07973	0.58817	6.23489

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR2X1	A	0.00000	0.00000	0.00000
	A	1.85506	1.87348	2.18076
	B	0.00000	0.00000	0.00000
	B	2.93878	2.95187	3.20602

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR2X1	A	0.00000	0.00000	0.00000
	A	1.25420	1.26154	1.47467
	B	0.00000	0.00000	0.00000
	B	2.95762	2.97251	3.23968

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.04884	0.04743	0.04726

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	3.12398	3.12324	3.12350

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(A * Y)	0.00000	0.00000	0.00000
	(A * Y)	0.01973	0.01973	0.01974

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR2X1	(A * Y)	0.00000	0.00000	0.00000
	(A * Y)	1.84555	1.84548	1.84610

OR3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	0	0	0
0	x	1	1
x	1	x	1
1	x	x	1

Footprint

Cell Name	Area
OR3X1	62.15760

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
OR3X1	0.01077	0.01055	0.01054	5.59917

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
OR3X1	0.00000	1.61262	5.01932

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR3X1	A->Y (RR)	0.06011	0.57707	6.24282
	B->Y (RR)	0.05969	0.59665	6.24269
	C->Y (RR)	0.05561	0.61035	6.57160

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
OR3X1	A->Y (FF)	0.14587	0.65536	6.33036
	B->Y (FF)	0.13284	0.66159	6.37101
	C->Y (FF)	0.10810	0.64477	6.36220

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR3X1	A	0.00000	0.00000	0.00000
	A	1.89623	1.91244	2.20987
	B	0.00000	0.00000	0.00000
	B	1.61743	1.62788	1.84353
	C	0.00000	0.00000	0.00000
	C	2.55377	2.56333	2.75602

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
OR3X1	A	0.00000	0.00000	0.00000
	A	0.62670	0.63086	0.80224
	B	0.00000	0.00000	0.00000
	B	1.31344	1.31873	1.48721
	C	0.00000	0.00000	0.00000
	C	2.58839	2.59926	2.81567

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.01815	0.01744	0.01735
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.06575	0.06442	0.06428
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	0.04924	0.04754	0.04731

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.06099	0.06106	0.06113
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	1.53524	1.53526	1.53528
	(!B * C * Y)	0.00000	0.00000	0.00000
	(!B * C * Y)	2.67809	2.67732	2.67786

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.02082	0.02084	0.02087
	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	0.03902	0.03901	0.03900
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	0.03893	0.03729	0.03698

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.06468	0.06470	0.06470
	(A * !C * Y)	0.00000	0.00000	0.00000
	(A * !C * Y)	1.87996	1.87987	1.88012
	(!A * C * Y)	0.00000	0.00000	0.00000
	(!A * C * Y)	2.75154	2.75082	2.75146

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.01258	0.01257	0.01256
	(A * !B * Y)	0.00000	0.00000	0.00000
	(A * !B * Y)	0.01730	0.01729	0.01728

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
OR3X1	(B * Y)	0.00000	0.00000	0.00000
	(B * Y)	0.08817	0.08816	0.08819
	(A * !B * Y)	0.00000	0.00000	0.00000
	(A * !B * Y)	1.88337	1.88336	1.88369

TIEHI

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Footprint

Cell Name	Area
TIEHI	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	Y
TIEHI	12.24737

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TIEHI	0.00000	0.00000	0.00000

TIELO

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage 1.80,
Temp 25.00*

Footprint

Cell Name	Area
TIELO	24.83200

Pin Capacitance Information

Cell Name	Max Cap(pf)
	YN
TIELO	26.69489

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TIELO	0.00000	0.00000	0.00000

TMRDFFQNX1

sky130_rhbd_tt_1P8_25C.ccs Cell
Library: Process , Voltage 1.80, Temp
25.00

Truth Table

INPUT		OUTPUT
D	CLK	QN
0	R	1
1	R	0
x	x	IQN

Footprint

Cell Name	Area
TMRDFFQNX1	584.71600

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	QN
TMRDFFQNX1	0.04012	0.07454	3.49318

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFQNX1	0.00000	74.65890	102.09600

Delay Information

Delay(ns) to QN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQNX1	CLK->QN (RR)	0.37922	1.41564	10.87570

Delay(ns) to QN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQNX1	CLK->QN (RF)	0.22685	0.72048	4.39873

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQNX1	hold	CLK (R)	0.04271	0.07129	0.59435
	setup	CLK (R)	0.09551	0.14461	0.68509

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQNX1	hold	CLK (R)	-0.03492	-0.08330	-0.56799
	setup	CLK (R)	0.04934	0.10599	1.02331

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQNX1	min_pulse_width	CLK ()	D	0.14159	0.67383	16.50020
	min_pulse_width	CLK ()	!D	0.14411	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQNX1	min_pulse_width	CLK ()	D	0.17942	0.67383	16.50020
	min_pulse_width	CLK ()	!D	0.09366	0.67383	16.50020

Power Information

Internal switching power(pJ) to QN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	76.28210	76.35650	77.67420

Internal switching power(pJ) to QN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQNX1	CLK	0.00000	0.00000	0.00000
	CLK	63432.30000	63452.60000	63470.10000

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	72626.40000	72623.60000	72627.20000
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	72289.90000	72290.50000	72300.90000
	!CLK	0.00000	0.00000	0.00000
	!CLK	71948.40000	71948.50000	71948.80000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(CLK * QN)	0.00000	0.00000	0.00000
	(CLK * QN)	63911.70000	63911.90000	63913.90000
	(CLK * !QN)	0.00000	0.00000	0.00000
	(CLK * !QN)	64606.70000	64607.40000	64610.40000
	!CLK	0.00000	0.00000	0.00000
	!CLK	89655.30000	89657.50000	89666.10000

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	65524.70000	66210.00000	66677.60000
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	76.35800	76.42560	77.69220

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQNX1	(D * QN)	0.00000	0.00000	0.00000
	(D * QN)	102986.00000	103926.00000	100262.00000
	(D * !QN)	0.00000	0.00000	0.00000
	(D * !QN)	97597.30000	97589.80000	97663.60000
	(!D * QN)	0.00000	0.00000	0.00000
	(!D * QN)	98685.20000	98667.50000	98691.20000
	(!D * !QN)	0.00000	0.00000	0.00000
	(!D * !QN)	88931.90000	88980.40000	89129.10000

TMRDFFQX1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
D	CLK	Q
0	R	0
1	R	1
x	x	IQ

Footprint

Cell Name	Area
TMRDFFQX1	601.94318

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	D	CLK	Q
TMRDFFQX1	0.04023	0.07454	5.60831

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFQX1	0.00000	76.52610	108.74900

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQX1	CLK->Q (RR)	0.26459	0.90873	7.28733

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFQX1	CLK->Q (RF)	0.44315	0.97252	5.82538

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQX1	hold	CLK (R)	0.04202	0.06642	0.56217
	setup	CLK (R)	0.09267	0.13542	0.66855

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFQX1	hold	CLK (R)	-0.03535	-0.08330	-0.56424
	setup	CLK (R)	0.04910	0.10497	1.01514

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQX1	min_pulse_width	CLK ()	D	0.14411	0.67383	16.50020
	min_pulse_width	CLK ()	!D	0.14411	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFQX1	min_pulse_width	CLK ()	D	0.17438	0.67383	16.50020
	min_pulse_width	CLK ()	!D	0.09366	0.67383	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	63396.40000	63414.40000	63429.70000

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFQX1	CLK	0.00000	0.00000	0.00000
	CLK	80.66200	80.73100	82.01210

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	72260.50000	72261.00000	72270.00000
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	72253.20000	72251.50000	72255.50000
	!CLK	0.00000	0.00000	0.00000
	!CLK	71573.20000	71570.40000	71570.90000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(CLK * Q)	0.00000	0.00000	0.00000
	(CLK * Q)	64324.00000	64326.40000	64328.70000
	(CLK * !Q)	0.00000	0.00000	0.00000
	(CLK * !Q)	63782.70000	63783.50000	63784.80000
	!CLK	0.00000	0.00000	0.00000
	!CLK	87187.00000	87173.20000	87190.70000

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	65806.70000	66456.60000	66896.10000
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	80.63480	80.69970	81.96350

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFQX1	(D * Q)	0.00000	0.00000	0.00000
	(D * Q)	98360.20000	98483.20000	98698.30000
	(D * !Q)	0.00000	0.00000	0.00000
	(D * !Q)	103202.00000	103983.00000	100166.00000
	(!D * Q)	0.00000	0.00000	0.00000
	(!D * Q)	88914.20000	88945.70000	89124.40000
	(!D * !Q)	0.00000	0.00000	0.00000
	(!D * !Q)	99446.70000	99427.30000	99442.40000

TMRDFFRNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell
Library: Process , Voltage 1.80,
Temp 25.00

Truth Table

INPUT			OUTPUT
D	RN	CLK	Q
x	x	x	-

Footprint

Cell Name	Area
TMRDFFRNQX1	705.30640

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	RN	CLK	Q
TMRDFFRNQX1	0.04039	0.09153	0.06859	0.50970

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFRNQX1	0.00000	762910.00000	1055710.00000

Delay Information

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFRNQX1	CLK->Q (FF)	522.64300	524.94600	527.66700
	D->Q (RF)	0.73216	1.19399	5.05947

Constraint Information

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFRNQX1	min_pulse_width	CLK ()	(!D * RN)	0.34653	0.73086	16.50020

Power Information

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFRNQX1	CLK	-0.01412	-0.16195	-0.82383
	CLK	531094.00000	531095.00000	531112.00000
	D	-0.01414	-0.16218	-0.82496
	D	473366.00000	473365.00000	473367.00000

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFRNQX1	(CLK * RN * Q)	0.00000	0.00000	0.00000
	(CLK * RN * Q)	427305.00000	427304.00000	427301.00000
	(!CLK * RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * RN * !Q)	473834.00000	473835.00000	473835.00000

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFRNQX1	(!CLK * RN * !Q)	0.00000	0.00000	0.00000
	(!CLK * RN * !Q)	531382.00000	531382.00000	531385.00000

TMRDFFSNQX1

sky130_rhbd_tt_1P8_25C.ccs Cell
Library: Process , Voltage 1.80,
Temp 25.00

Truth Table

INPUT			OUTPUT
D	SN	CLK	Q
0	1	R	0
1	1	R	1
x	0	x	1
x	1	x	IQ

Footprint

Cell Name	Area
TMRDFFSNQX1	670.85199

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	D	SN	CLK	Q
TMRDFFSNQX1	0.04118	0.07158	0.07187	5.34521

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
TMRDFFSNQX1	0.00000	86.97680	149.96700

Delay Information

Delay(ns) to Q rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQX1	CLK->Q (RR)	0.23650	0.87765	7.11499
	SN->Q (FR)	0.15186	0.87743	8.00698

Delay(ns) to Q falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
TMRDFFSNQX1	CLK->Q (RF)	0.46243	0.97644	5.61089

Constraint Information

Constraints(ns) for D rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	hold	CLK (R)	0.01169	0.01620	0.49654
	setup	CLK (R)	0.12240	0.17651	0.99303

Constraints(ns) for D falling :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	hold	CLK (R)	-0.07279	-0.15399	-1.14942
	setup	CLK (R)	0.09258	0.17976	1.49071

Constraints(ns) for D rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	hold	CLK (R)	SN	0.01169	0.01620	0.49654
	setup	CLK (R)	SN	0.12240	0.17651	0.99303

Constraints(ns) for D falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	hold	CLK (R)	SN	-0.07279	-0.15399	-1.14942
	setup	CLK (R)	SN	0.09258	0.17976	1.49071

Constraints(ns) for SN rising :

Cell Name	Timing Check	Ref Pin(trans)	Reference Slew Rate(ns)		
			first	mid	last
TMRDFFSNQX1	recovery	CLK (R)	0.02565	0.02435	3.86768
	removal	CLK (R)	-0.01690	-0.01262	-0.11857

Constraints(ns) for SN rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	recovery	CLK (R)	!D	0.02565	0.02435	3.86768
	removal	CLK (R)	!D	-0.01690	-0.01262	-0.11857

Constraints(ns) for SN falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	SN ()	(CLK * D)	0.09618	0.67383	16.50020
	min_pulse_width	SN ()	(CLK * !D)	0.09618	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * D)	0.09871	0.67383	16.50020
	min_pulse_width	SN ()	(!CLK * !D)	0.09871	0.67383	16.50020

Constraints(ns) for CLK rising (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.13906	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.15168	0.67383	16.50020

Constraints(ns) for CLK falling (conditional):

Cell Name	Timing Check	Ref Pin(trans)	When	Reference Slew Rate(ns)		
				first	mid	last
TMRDFFSNQX1	min_pulse_width	CLK ()	(D * SN)	0.19708	0.67383	16.50020
	min_pulse_width	CLK ()	(!D * SN)	0.10375	0.67383	16.50020

Power Information

Internal switching power(pJ) to Q rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	58.62820	58.64290	59.30320
	SN	57.53720	57.57260	58.20930

Internal switching power(pJ) to Q falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	CLK	0.00000	0.00000	0.00000
	CLK	105.56300	105.57600	106.21400

Passive power(pJ) for D rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	100.05800	100.06300	100.06700
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	40.16600	40.17650	40.77090

Passive power(pJ) for D falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	0.00000	0.00000	0.00000
	$(CLK * SN) + (CLK * !SN * Q) + (!CLK * !SN * Q)$	101.96200	101.96700	101.97200
	$(!CLK * SN)$	0.00000	0.00000	0.00000
	$(!CLK * SN)$	54.53500	54.56290	55.20090

Passive power(pJ) for SN rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * Q) + (!CLK * D * Q)$	0.00000	0.00000	0.00000
	$(CLK * Q) + (!CLK * D * Q)$	55.18910	55.19740	55.20040
	$(!CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * Q)$	44.14950	44.16040	44.73120

Passive power(pJ) for SN falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(CLK * Q) + (!CLK * D * Q)$	0.00000	0.00000	0.00000
	$(CLK * Q) + (!CLK * D * Q)$	58.53210	58.53770	58.54070
	$(!CLK * !D * Q)$	0.00000	0.00000	0.00000
	$(!CLK * !D * Q)$	12.03790	12.06720	12.67370

Passive power(pJ) for CLK rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(D * SN) + (D * !SN * Q)$	0.00000	0.00000	0.00000
	$(D * SN) + (D * !SN * Q)$	59.21670	59.23170	59.86920
	$(!D * SN)$	0.00000	0.00000	0.00000
	$(!D * SN)$	104.87900	104.88400	105.49100
	$(!D * !SN * Q)$	0.00000	0.00000	0.00000
	$(!D * !SN * Q)$	33.00840	33.05580	33.83370

Passive power(pJ) for CLK falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
TMRDFFSNQX1	$(D * SN) + (D * !SN * Q)$	0.00000	0.00000	0.00000
	$(D * SN) + (D * !SN * Q)$	40.88080	40.89440	41.58210
	$(!D * SN)$	0.00000	0.00000	0.00000
	$(!D * SN)$	53.50670	53.52170	54.14210
	$(!D * !SN * Q)$	0.00000	0.00000	0.00000
	$(!D * !SN * Q)$	11.54500	11.57570	12.43390

VOTER3X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT			OUTPUT
A	B	C	Y
0	0	x	0
x	1	x	1
1	0	0	0
1	x	1	1

Footprint

Cell Name	Area
VOTER3X1	102.35440

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	Y
VOTER3X1	0.02219	0.02088	0.02070	5.44865

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
VOTER3X1	0.00000	3.89005	10.36970

Delay Information

Delay(ns) to Y rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTER3X1	A->Y (RR)	0.08406	0.68135	6.73295
	B->Y (RR)	0.09325	0.66129	6.43137
	C->Y (RR)	0.09018	0.67295	6.68167

Delay(ns) to Y falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTER3X1	A->Y (FF)	0.14296	0.69506	6.19532
	B->Y (FF)	0.20123	0.81270	6.67688
	C->Y (FF)	0.17747	0.72865	6.38445

Power Information

Internal switching power(pJ) to Y rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTER3X1	A	5.10375	5.11049	5.28329
	B	1.62656	1.63119	1.78992
	C	5.10184	5.10593	5.27162

Internal switching power(pJ) to Y falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTER3X1	A	-0.01415	-0.67262	-8.82681
	A	2.09455	2.10092	2.29680
	B	-0.01415	-0.67262	-8.82681
	B	1.16497	1.16860	1.31089
	C	-0.01415	-0.67262	-8.82681
	C	2.10282	2.10884	2.31999

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	0.01566	0.01542	0.01540
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	0.11434	0.11259	0.11201
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	1.45968	1.45909	1.45905

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(B * C * Y)	0.00000	0.00000	0.00000
	(B * C * Y)	1.64146	1.64151	1.64166
	(B * !C * Y)	0.00000	0.00000	0.00000
	(B * !C * Y)	1868.73000	1844.81000	1839.36000
	(!B * !C * !Y)	0.00000	0.00000	0.00000
	(!B * !C * !Y)	2.22151	2.22176	2.22183

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	0.02719	0.02411	0.02353
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	3.99749	4.07056	4.15239

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * C * Y)	0.00000	0.00000	0.00000
	(A * C * Y)	5.84834	5.84801	5.84806
	(!A * !C * !Y)	0.00000	0.00000	0.00000
	(!A * !C * !Y)	2.21577	2.21416	2.21468

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * B * Y)	0.00000	0.00000	0.00000
	(A * B * Y)	0.01752	0.01749	0.01750
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	1.25835	1.25813	1.25824
	(!A * !B * !Y)	0.00000	0.00000	0.00000
	(!A * !B * !Y)	1.45749	1.45692	1.45627

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTER3X1	(A * B * Y)	0.00000	0.00000	0.00000
	(A * B * Y)	0.16843	0.16848	0.16851
	(!A * B * Y)	0.00000	0.00000	0.00000
	(!A * B * Y)	1761.37000	1741.11000	1735.17000
	(!A * !B * !Y)	0.00000	0.00000	0.00000
	(!A * !B * !Y)	2.22025	2.21901	2.21920

VOTERN3X1

sky130_rhbd_tt_1P8_25C.ccs Cell Library:
Process , Voltage 1.80, Temp 25.00

Truth Table

INPUT			OUTPUT
A	B	C	YN
0	0	x	1
0	1	0	1
x	1	1	0
1	0	0	1
1	x	1	0
1	1	x	0

Footprint

Cell Name	Area
VOTERN3X1	85.12720

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)
	A	B	C	YN
VOTERN3X1	0.02297	0.02197	0.02083	1.85378

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
VOTERN3X1	0.00000	2.49125	10.79090

Delay Information

Delay(ns) to YN rising :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTERN3X1	A->YN (FR)	0.13961	1.11685	9.75043
	B->YN (FR)	0.15501	1.11436	9.53579
	C->YN (FR)	0.11098	1.07096	9.79229

Delay(ns) to YN falling :

Cell Name	Timing Arc(Dir)	Delay(ns)		
		First	Mid	Last
VOTERN3X1	A->YN (RF)	0.05719	0.57050	5.21995
	B->YN (RF)	0.05428	0.53588	4.89989
	C->YN (RF)	0.05599	0.58346	5.43067

Power Information

Internal switching power(pJ) to YN rising :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTERN3X1	A	0.05355	0.05498	0.10637
	B	0.07277	0.07254	0.09925
	C	0.06128	0.06308	0.11722

Internal switching power(pJ) to YN falling :

Cell Name	Input	Power(pJ)		
		first	mid	last
VOTERN3X1	A	4.74061	4.74031	4.76621
	B	4.74075	4.73895	4.76105
	C	3.78485	3.78563	3.80464

Passive power(pJ) for A rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	0.01482	0.01463	0.01462
	(!B * !C * YN)	0.00000	0.00000	0.00000
	(!B * !C * YN)	0.00423	0.00421	0.00418

Passive power(pJ) for A falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(B * C * !YN)	0.00000	0.00000	0.00000
	(B * C * !YN)	3.52077	3.52078	3.52091
	(!B * !C * YN)	0.00000	0.00000	0.00000
	(!B * !C * YN)	0.02488	0.02496	0.02530

Passive power(pJ) for B rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	0.04278	0.03933	0.03853
	(!A * !C * YN)	0.00000	0.00000	0.00000
	(!A * !C * YN)	-0.00067	-0.00072	-0.00075

Passive power(pJ) for B falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * C * !YN)	0.00000	0.00000	0.00000
	(A * C * !YN)	6.24353	6.24294	6.24259
	(!A * !C * YN)	0.00000	0.00000	0.00000
	(!A * !C * YN)	0.02484	0.02398	0.02389

Passive power(pJ) for C rising (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * B * !YN)	0.00000	0.00000	0.00000
	(A * B * !YN)	0.01452	0.01452	0.01451
	(!A * !B * YN)	0.00000	0.00000	0.00000
	(!A * !B * YN)	0.00200	0.00192	0.00193

Passive power(pJ) for C falling (conditional):

Cell Name	When	Power(pJ)		
		first	mid	last
VOTERN3X1	(A * B * !YN)	0.00000	0.00000	0.00000
	(A * B * !YN)	4.74559	4.74516	4.74575
	(!A * !B * YN)	0.00000	0.00000	0.00000
	(!A * !B * YN)	0.02467	0.02440	0.02434

XNOR2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process ,
Voltage 1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Footprint

Cell Name	Area
XNOR2X1	93.74080

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
XNOR2X1	0.02194	0.02438	2.53791

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
XNOR2X1	0.00000	9.81263	18.16860

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XNOR2X1	A->Y (RR)	B	0.08495	0.76227	6.27048
	A->Y (FR)	!B	0.07482	0.98510	9.94817
	B->Y (RR)	A	0.10345	0.82223	6.55198
	B->Y (FR)	!A	0.06325	0.95422	9.79971

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XNOR2X1	A->Y (FF)	B	0.07705	0.55353	4.49818
	A->Y (RF)	!B	0.04996	0.61688	6.30165
	B->Y (FF)	A	0.08940	0.56959	4.50900
	B->Y (RF)	!A	0.04494	0.60734	6.28031

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XNOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	5.00145	5.01529	5.27948
	A	!B	0.00000	0.00000	0.00000
	A	!B	0.07230	0.09021	0.37716
	B	A	0.00000	0.00000	0.00000
	B	A	1.83490	1.84384	2.05934
	B	!A	0.00000	0.00000	0.00000
	B	!A	0.06909	0.08861	0.40686

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XNOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	11.02960	11.04580	11.30350
	A	!B	0.00000	0.00000	0.00000
	A	!B	8.94399	8.95539	9.18979
	B	A	0.00000	0.00000	0.00000
	B	A	8.99027	9.00472	9.26470
	B	!A	0.00000	0.00000	0.00000
	B	!A	9.34972	9.35776	9.58167

XOR2X1

*sky130_rhbd_tt_1P8_25C.ccs Cell Library: Process , Voltage
1.80, Temp 25.00*

Truth Table

INPUT		OUTPUT
A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

Footprint

Cell Name	Area
XOR2X1	93.74080

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	A	B	Y
XOR2X1	0.02211	0.02441	2.57984

Leakage Information

Cell Name	Leakage(nW)		
	Min.	Avg	Max.
XOR2X1	0.00000	9.77674	19.21570

Delay Information

Delay(ns) to Y rising (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XOR2X1	A->Y (RR)	!B	0.08513	0.76760	6.33970
	A->Y (FR)	B	0.07698	0.98880	10.00830
	B->Y (RR)	!A	0.10631	0.78376	6.34887
	B->Y (FR)	A	0.06813	0.97652	9.98765

Delay(ns) to Y falling (conditional):

Cell Name	Timing Arc(Dir)	When	Delay(ns)		
			First	Mid	Last
XOR2X1	A->Y (FF)	!B	0.07677	0.56094	4.61292
	A->Y (RF)	B	0.04615	0.62231	6.39500
	B->Y (FF)	!A	0.08485	0.58665	4.59993
	B->Y (RF)	A	0.04046	0.64894	6.81881

Power Information

Internal switching power(pJ) to Y rising (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	0.54014	0.55538	0.81325
	A	!B	0.00000	0.00000	0.00000
	A	!B	1.73166	1.74551	2.01206
	B	A	0.00000	0.00000	0.00000
	B	A	0.53083	0.54532	0.79763
	B	!A	0.00000	0.00000	0.00000
	B	!A	1.74476	1.75476	2.02172

Internal switching power(pJ) to Y falling (conditional):

Cell Name	Input	When	Power(pJ)		
			first	mid	last
XOR2X1	A	B	0.00000	0.00000	0.00000
	A	B	15.38110	15.39660	15.68920
	A	!B	0.00000	0.00000	0.00000
	A	!B	6.22762	6.24265	6.49538
	B	A	0.00000	0.00000	0.00000
	B	A	15.37910	15.39560	15.68900
	B	!A	0.00000	0.00000	0.00000
	B	!A	6.23311	6.24481	6.46674